INRA
SCIENCE & IMPACT

Stics

HORTIS Aquitaine
ARVALIS Institut du végétal
ITB Institut Technique de la Betterave
cea
Poitou Charentes
Montpellier SupAgro
CETIOM Centre technique interprofessionnel des oléagineux métropolitains
Ctifl Centre technique interprofessionnel des fruits et légumes
Cemagref
CIRAD
MINES ParisTech
AGROTRANSFERT RESSOURCES ET TERRITOIRES
IRSN INSTITUT DE RADIOPROTECTION ET DE SÛRETÉ NUCLÉAIRE

3. Crop development and growth

L. Strullu,
on benhalf of the EPS (STICS project team)

ALIMENTATION
AGRICULTURE
ENVIRONNEMENT

Modulo-STICS : Crop development and growth

1. Development

2. Growth :

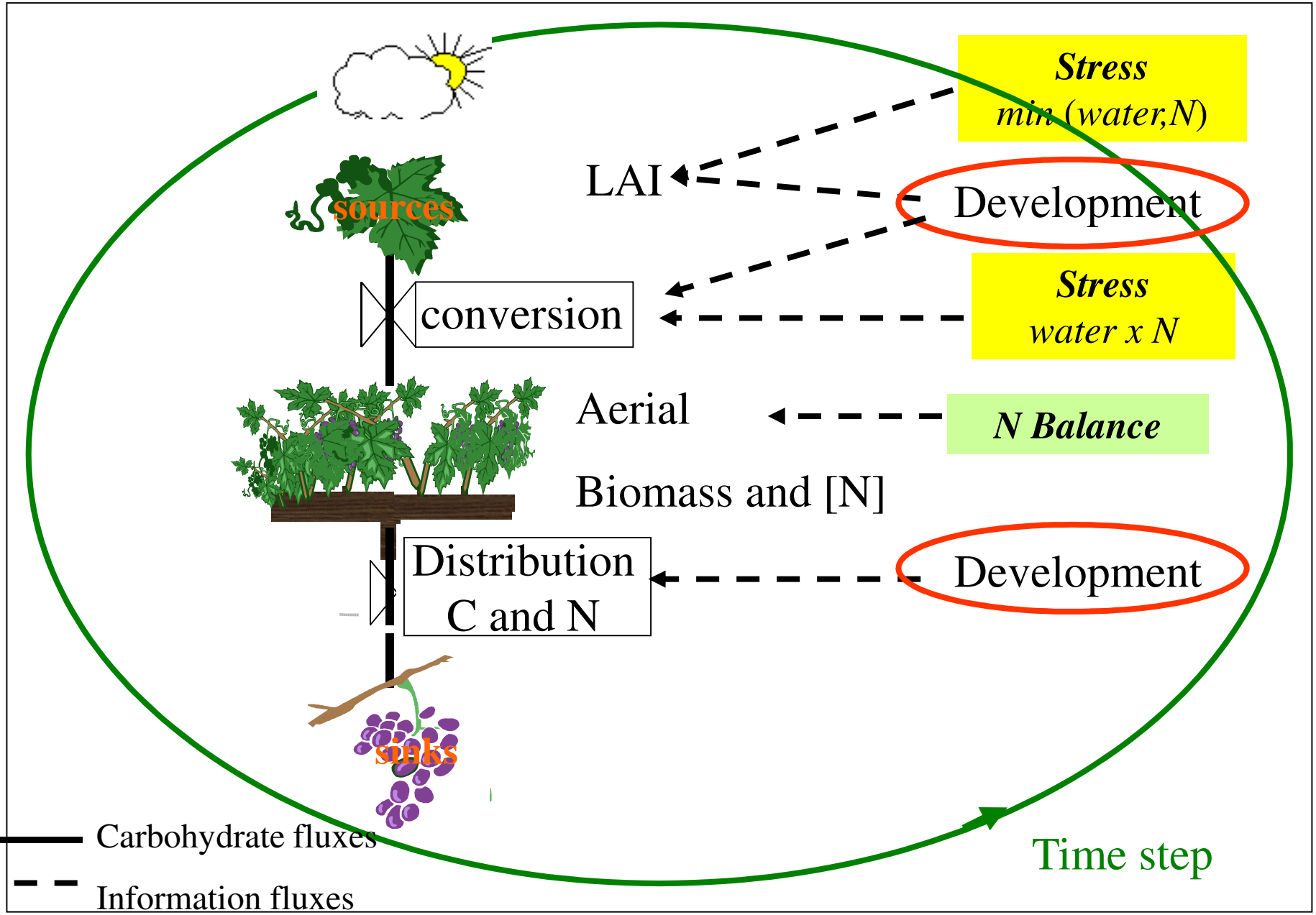
- Roots
- Leaf Area Index
- Radiations interception
- Biomass production
- N stress
- Water stress
- N fixation

3. Yield

4. Biomass distribution

Development

Development



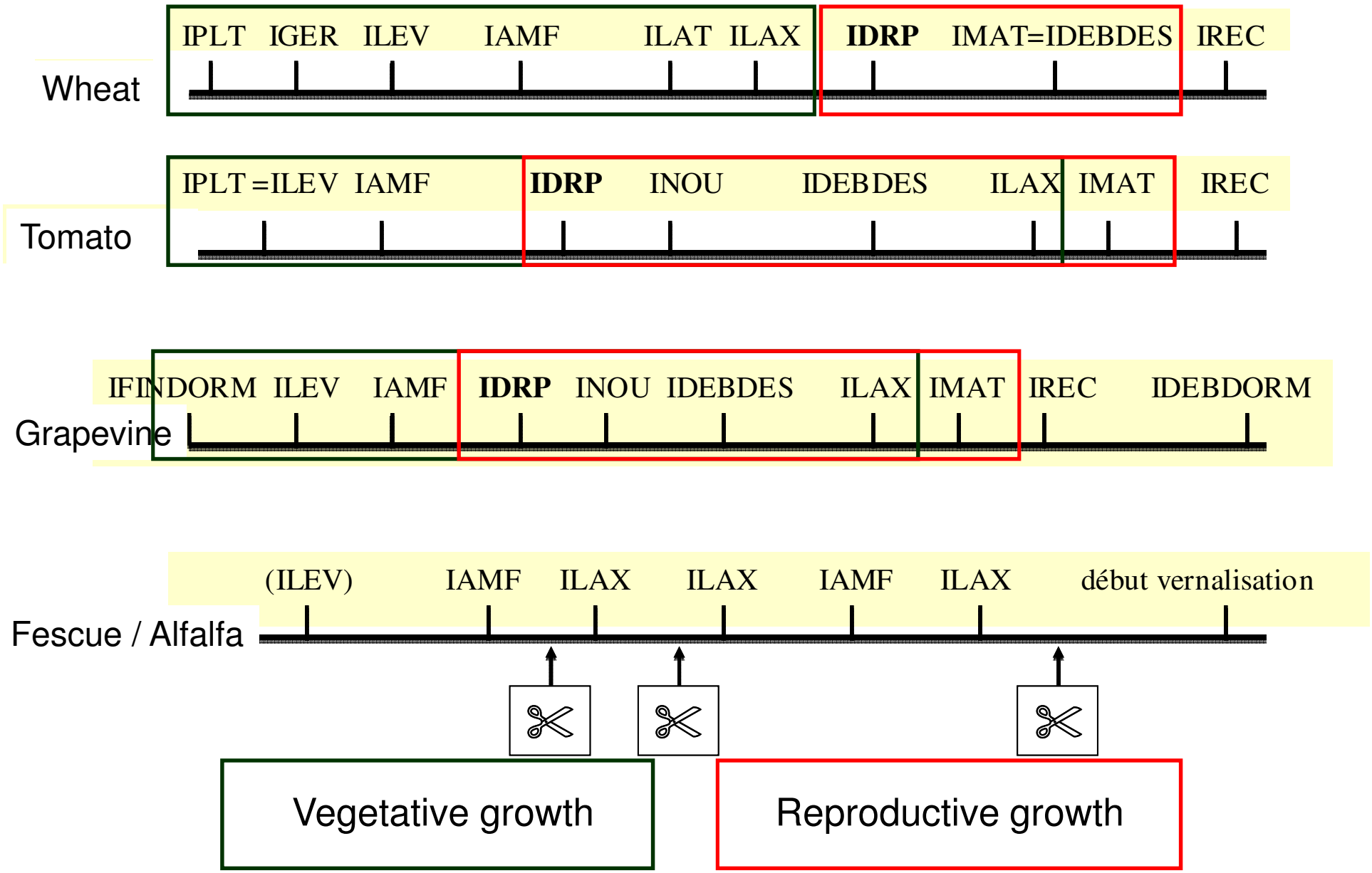
Development

Brisson et al., 2009

Vegetative stages (Leaf Area Index)	Reproductive stages (harvested organs filling)
IPLT: seedling	
IGER: germination	
IDEBDORM et IFINDORM: beginning and end of dormancy	
ILEV: Emergence for annual crops or Budbreak for perennial crops	
ILET: End of plantlet sensitivity to frost	ILAT: Beginning of the critical phase for grain number onset
IAMF: maximum acceleration of leaf growth, end of juvenile phase	IFLO: flowering (start of fruit sensitivity to frost)
	IDRP: Onset of filling of harvested organs
ILAX: End of LAI growth, emission of new Leaves stop	INOU: end of setting (indeterminate crops)
	IDEBDES: onset of water dynamics in fruits
	IMAT: physiological maturity
IREC: harvest	

Development

Brisson et al., 2009



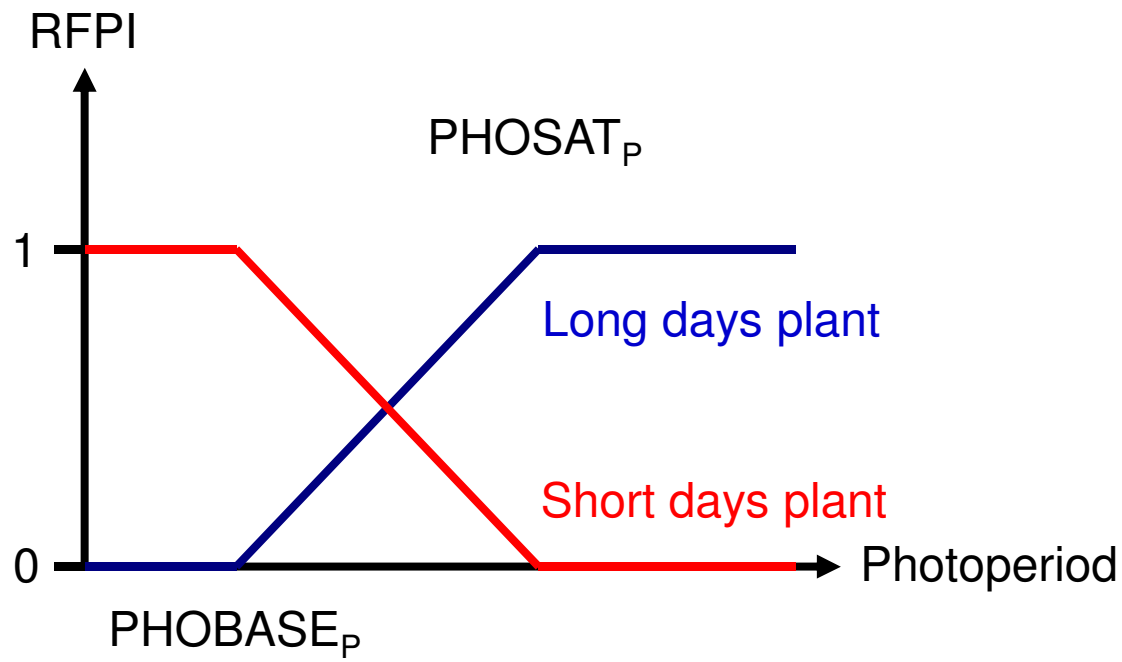
Development

Development unit of the crop (degree days) :

UDEVCULT(n) = Tmean(n) – Tmin

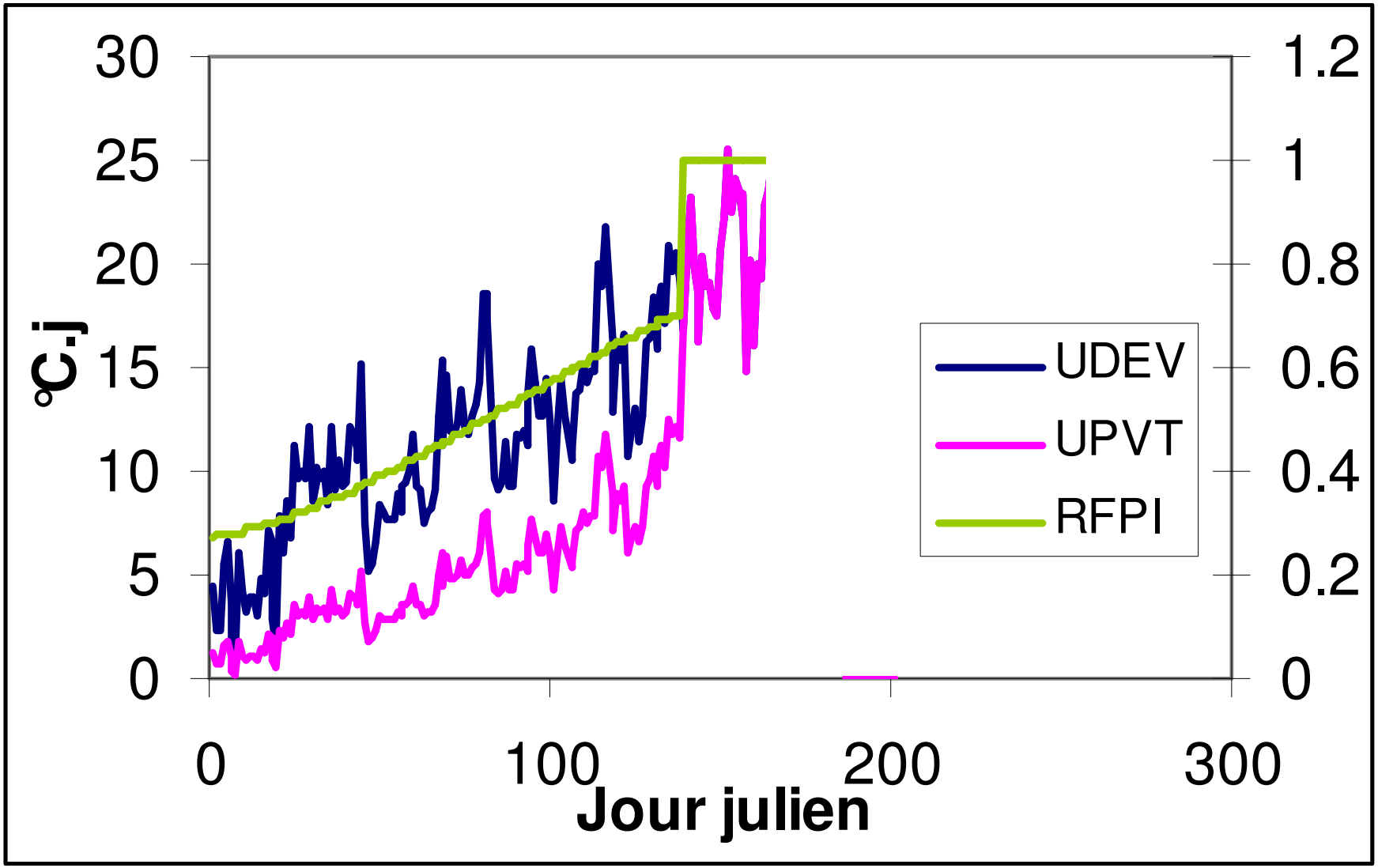
Real development unit of the crop, example for a photoperiodic plant :

UPVT(n) = UDEVCULT(n) * RFPI(n)

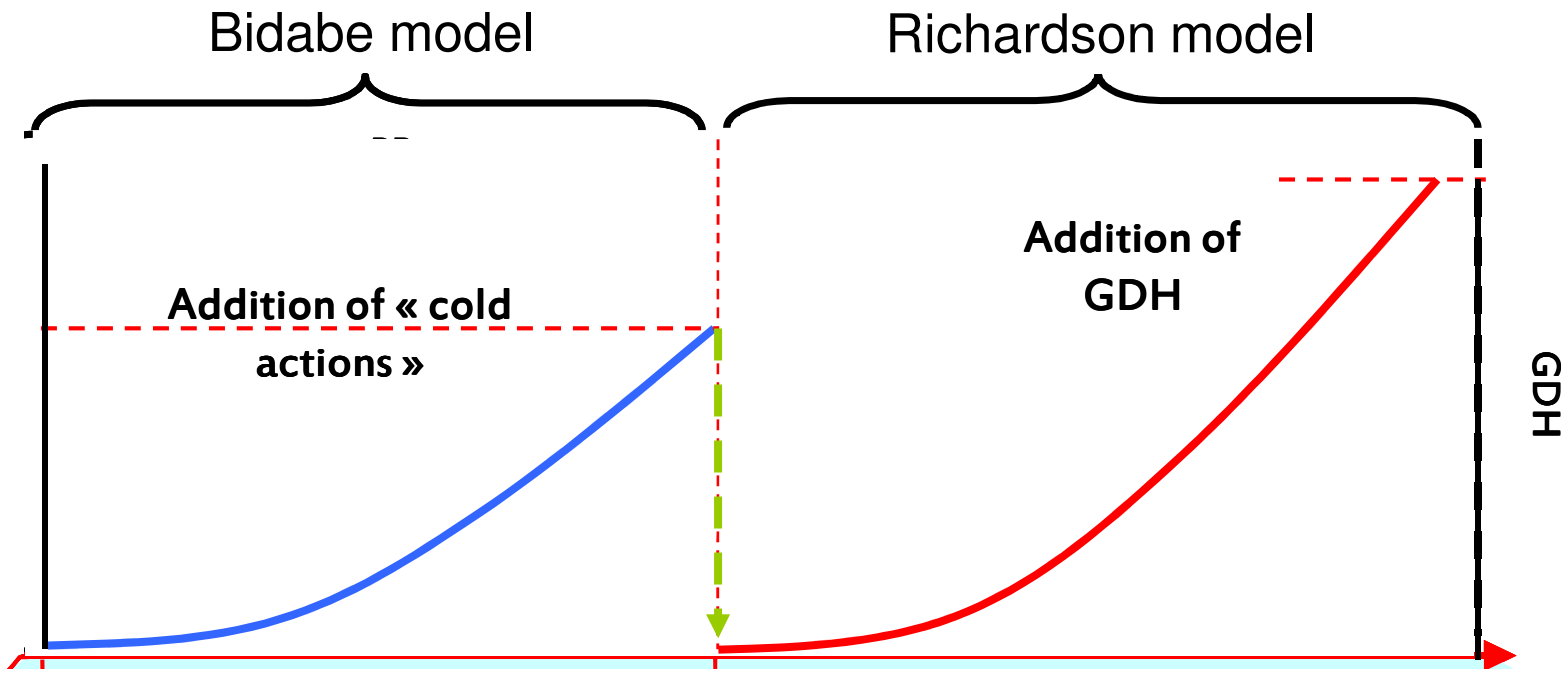


Development

Example : *Triticum durum* (long days plant)



Development : budbreak of perennial crops



Stage	Start of dormancy	End of dormancy	Budbreak
	DEBDORM	FINDORM	LEV

Garcia de Cortazar-Atauri et al., 2009

Development : Germination of annual crops

Humectation: oui ou non

- Si sol trop sec, la graine garde sa capacité germinative mais se n'humecte pas.
- Si humidité du sol ≥ seuil d'humidité exprimé en potentiel d'eau dans le sol (paramètre POTGERMI en MPa), la graine s'humecte.

Durée limite d'autotrophie de la graine

- Calcul d'un nombre de jours limite d'humectation NBJGRAUTO

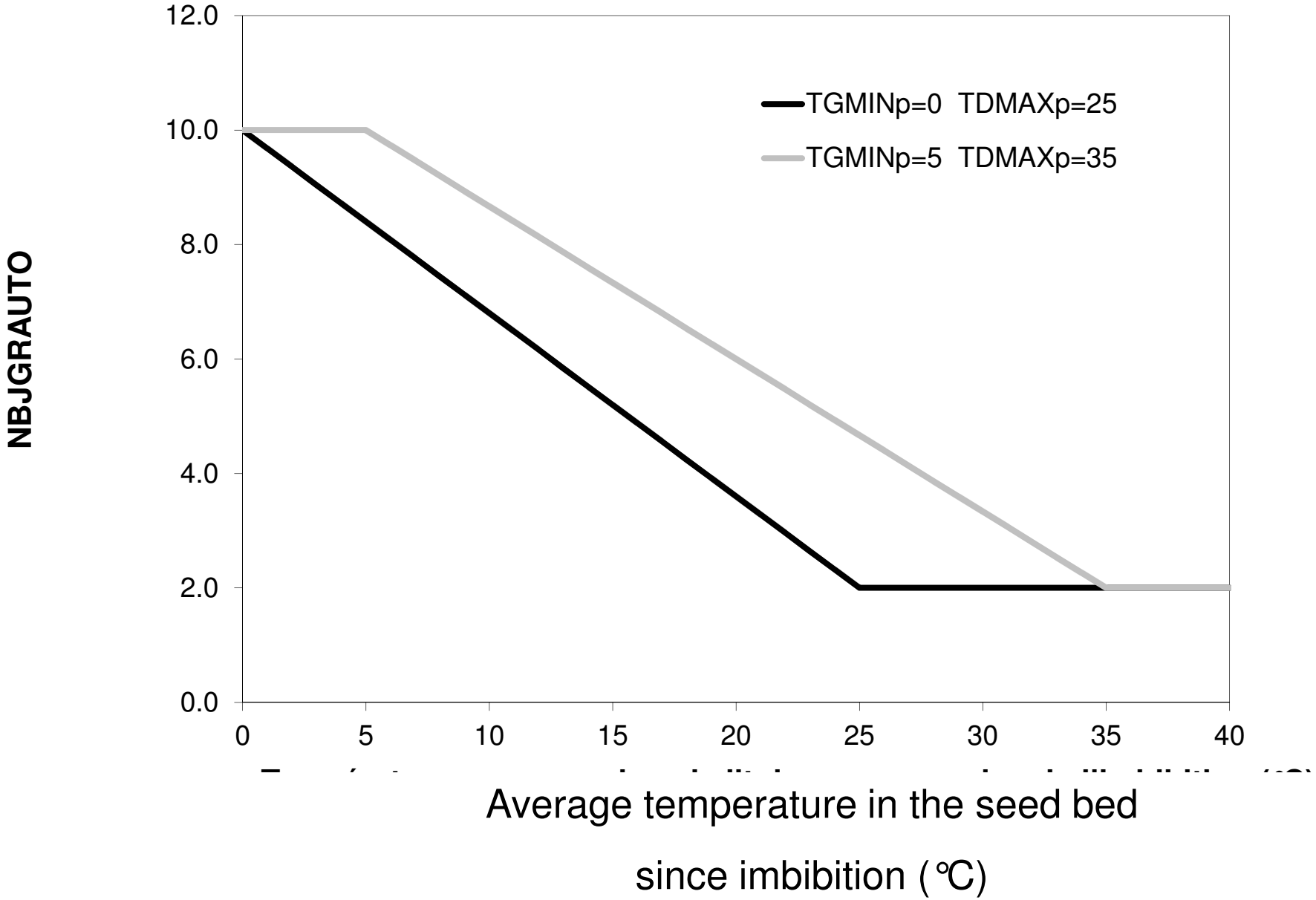
⇔ capacité d'autotrophie de la graine (fonction de la température dans le lit de semence, si faibles températures respiration et consommation des réserves sont moindre)

Densité de germination

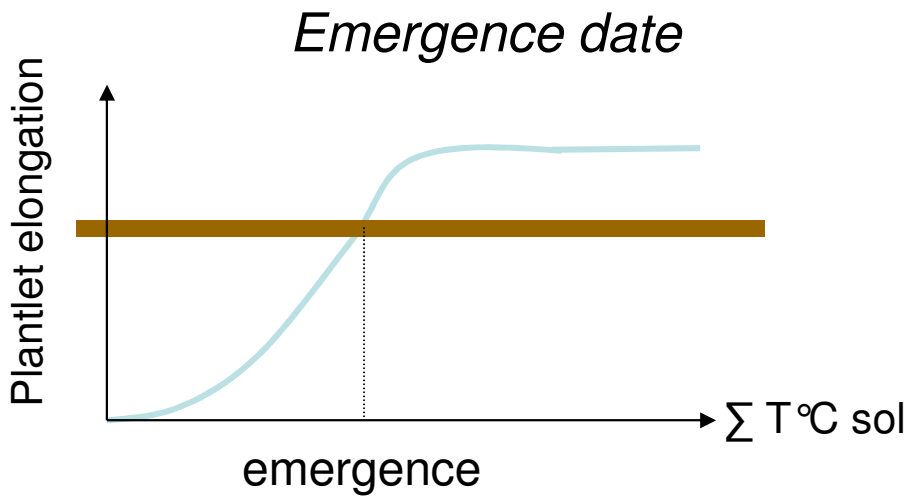
Si durée germination de la graine > nombre de jours NBJGRAUTO
la germination est déclenchée, mais avec des pertes en densité:

$$\text{DENSITEGER} = \text{DENSITE SEMIS} \times \frac{\text{SOMGER (somme de °C.j pendant NBJGRAUTO)}}{\text{STPLTGER}}$$

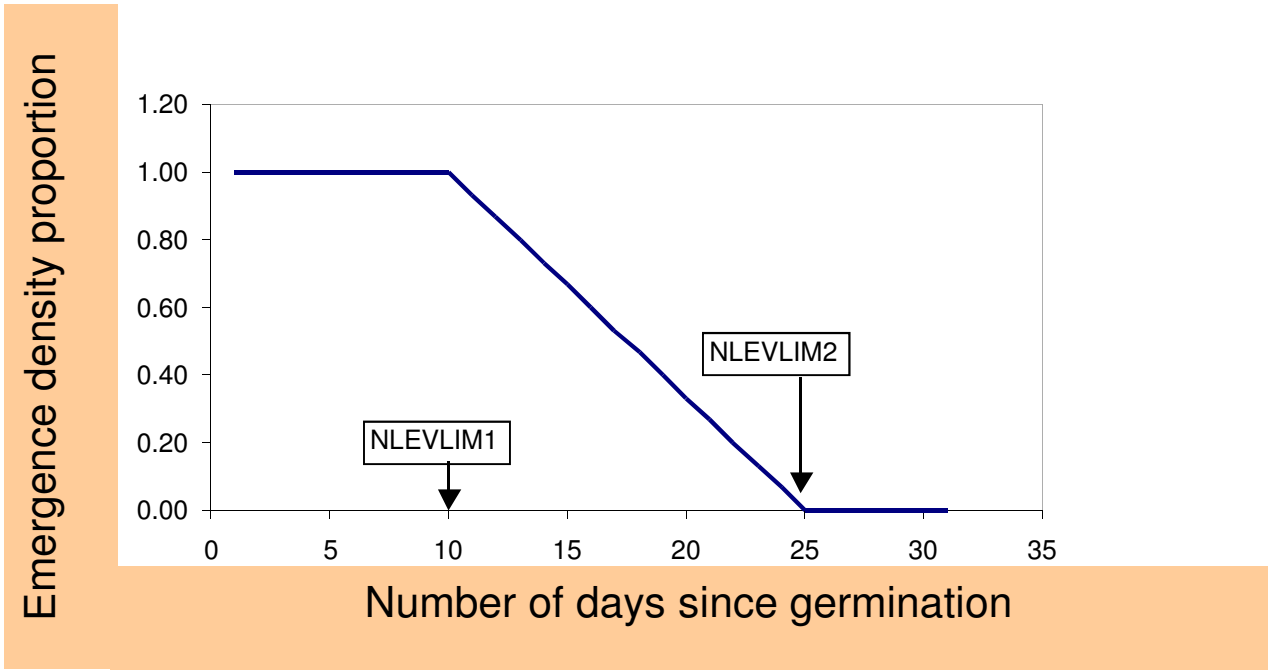
Development : Germination of annual crops



Development : Germination of annual crops



Decrease of emergence density : effect of soil surface crusting

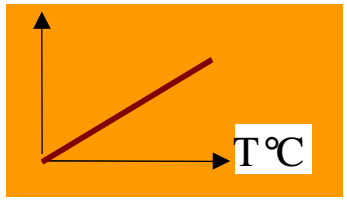


Roots

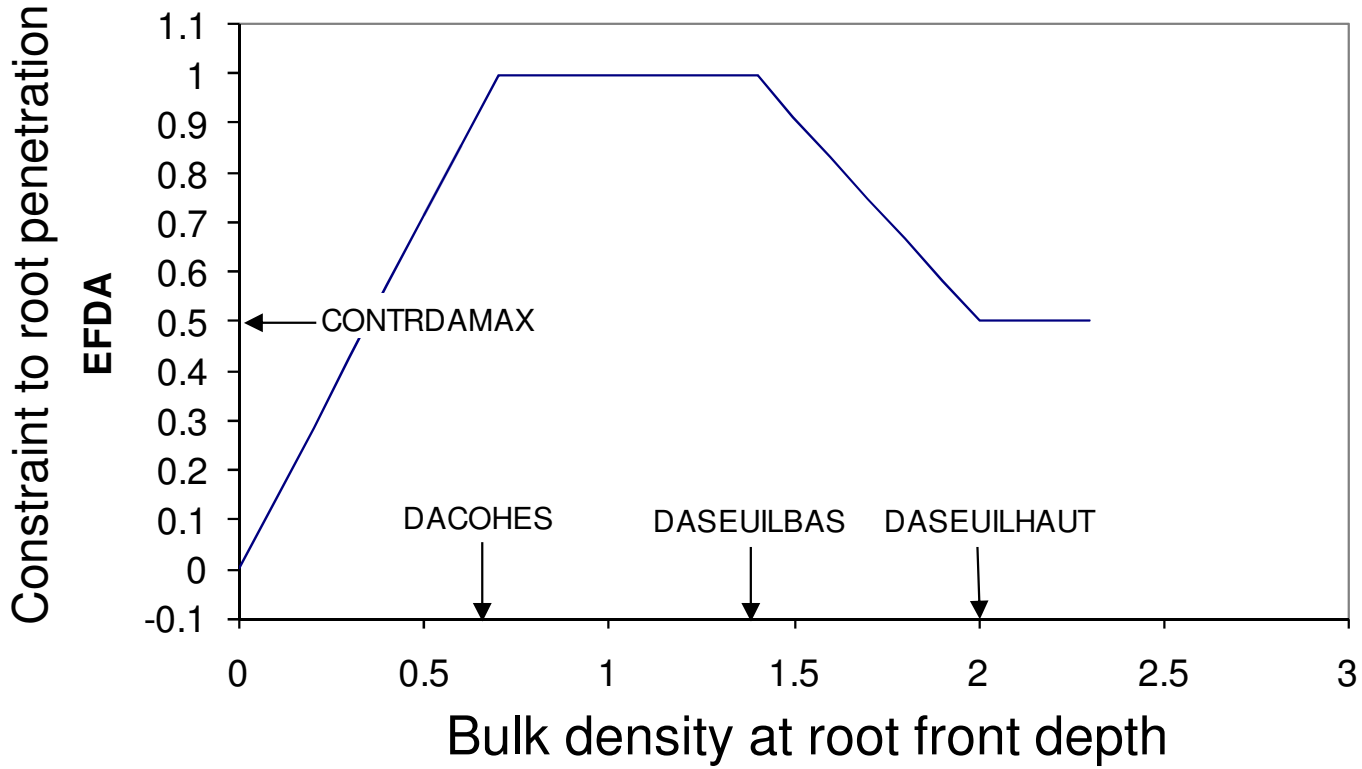
Growth : roots

Root front growth

Depth of root front

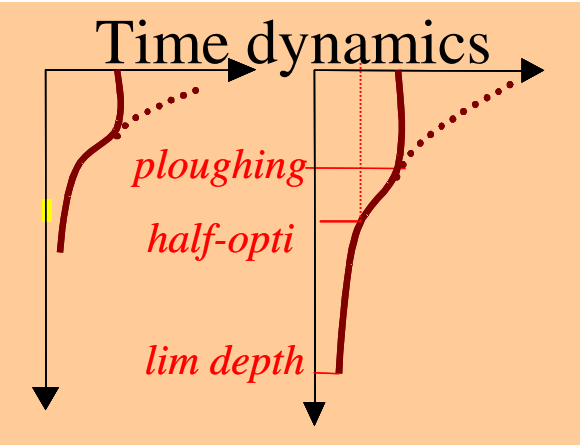


Water logging
Soil dryness
Bulk density

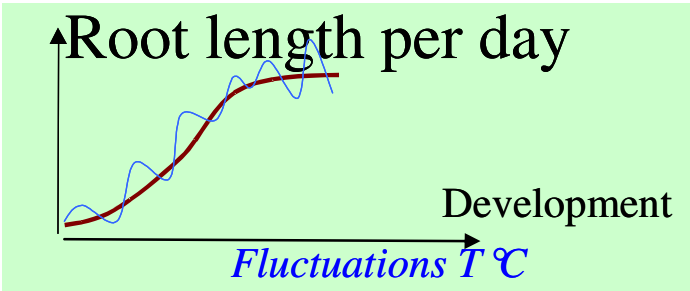


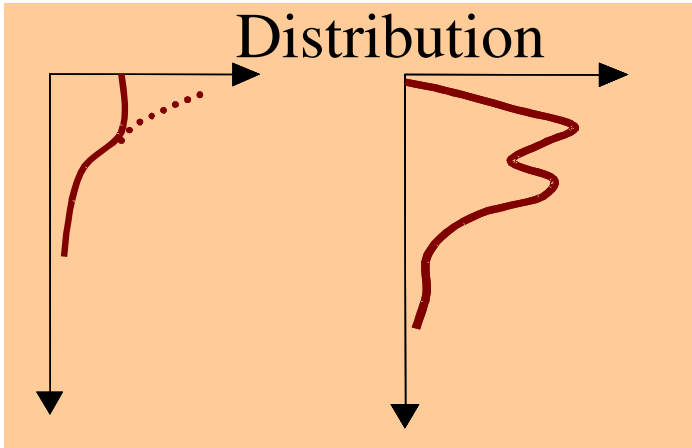
Growth : roots

Standard profile



True density

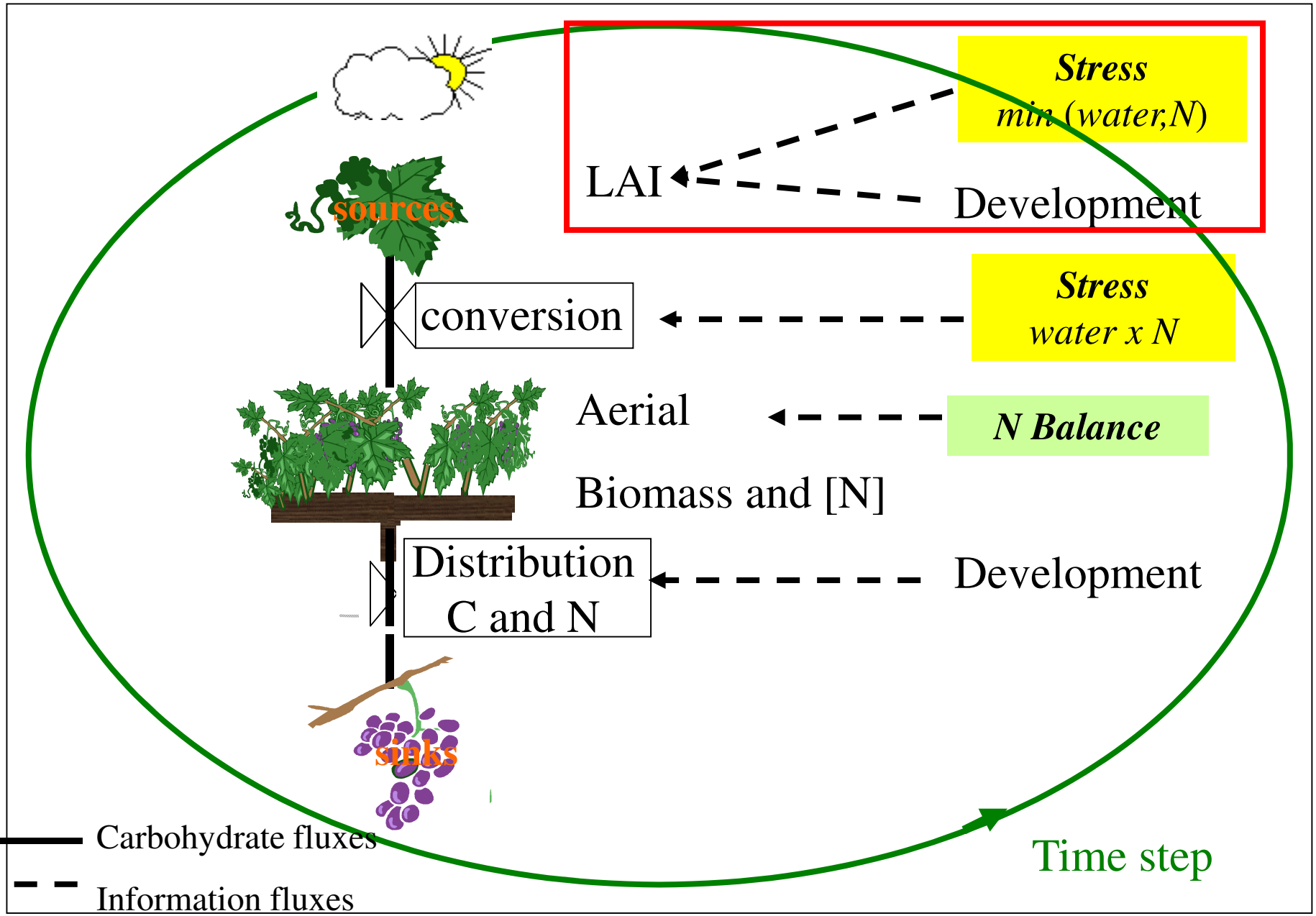




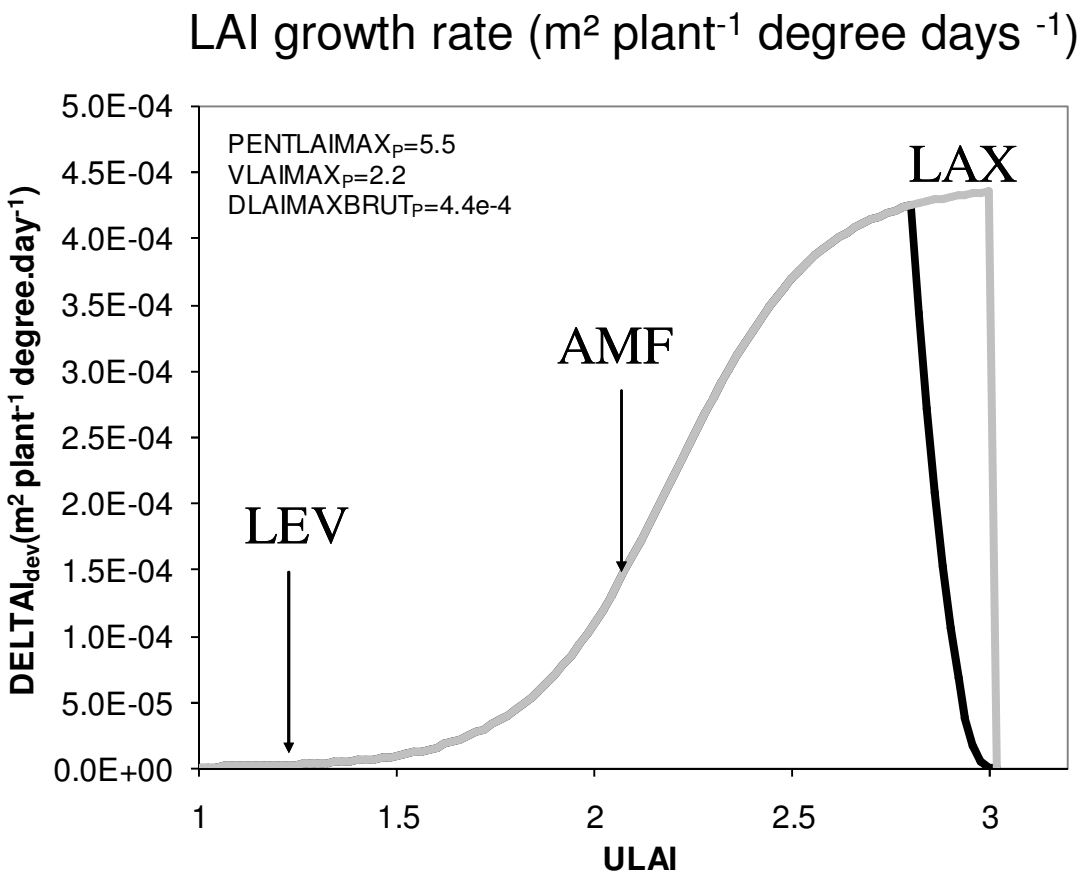
Effect of
- **Water logging**
- **Soil dryness**
- **Bulk density**

Leaf Area Index

Growth : Leaf area index



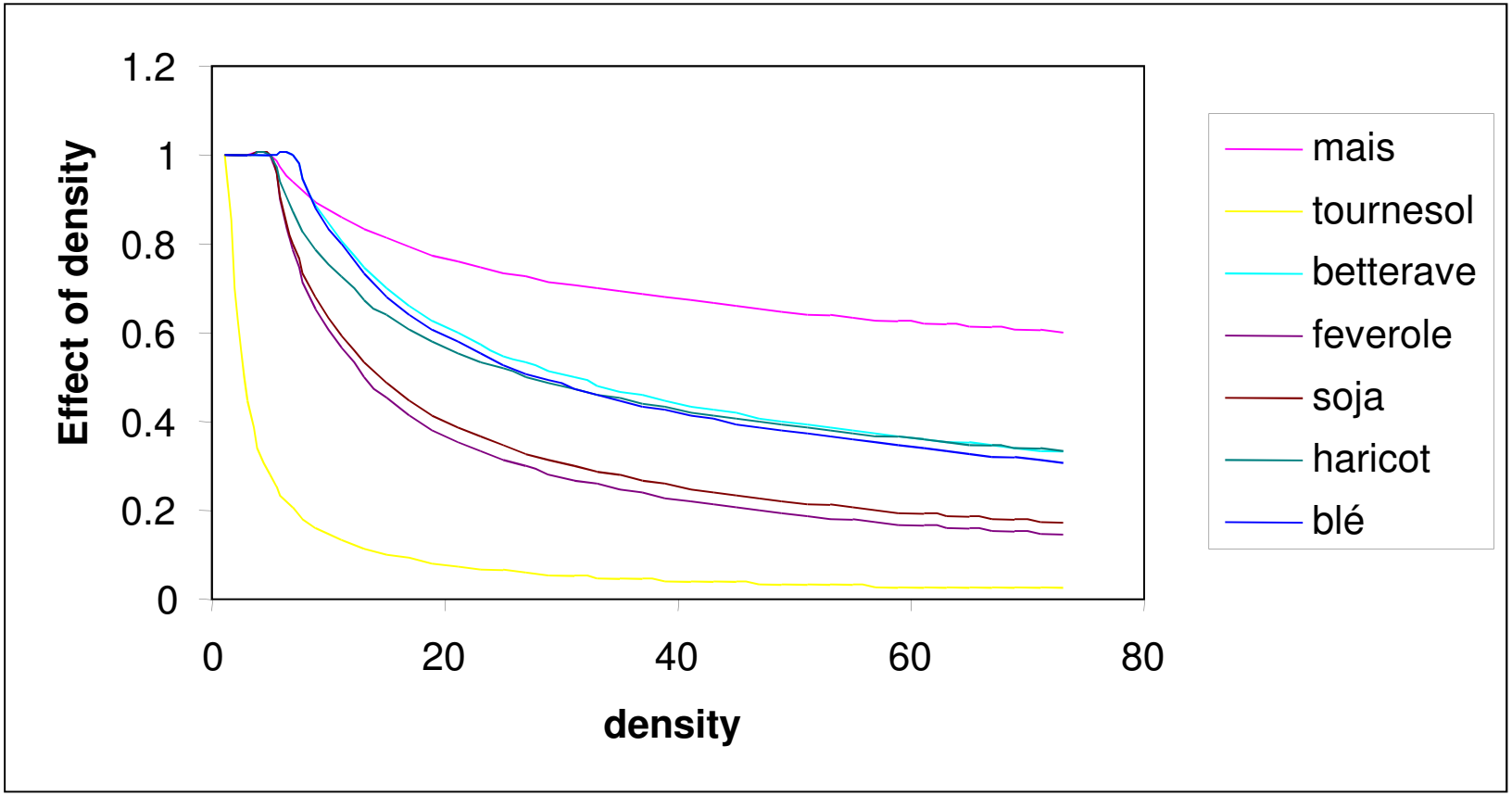
Growth : Leaf area index



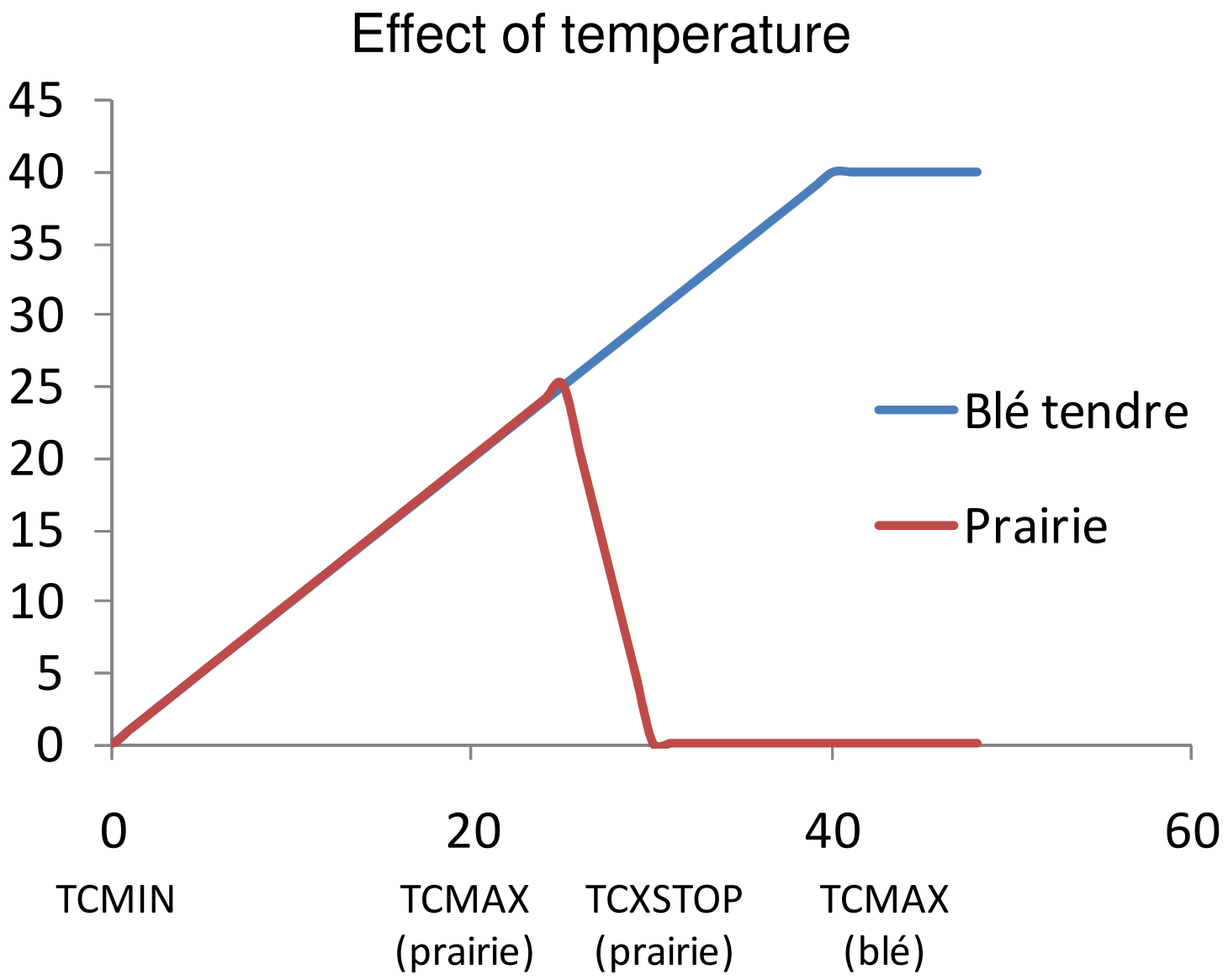
+ effects on potential growth :
- population density
- temperature
- stress H_2O, N
- trophic stress (indeterminate)
- senescence (if gross LAI)
- agricultural practices (cuts)

Growth : Leaf area index

Effect of plants density

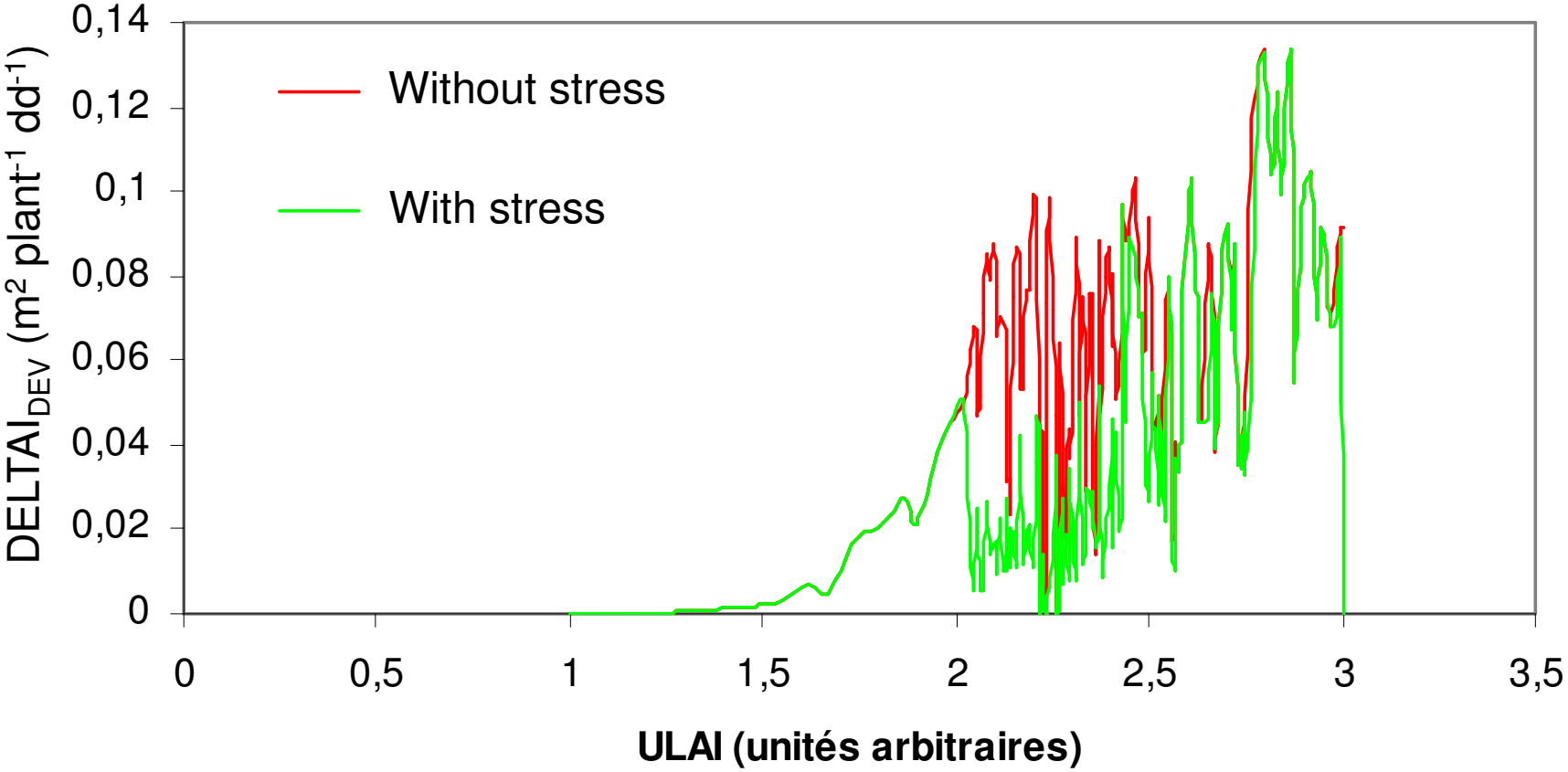


Growth : Leaf area index



Growth : Leaf area index

Effect of N and/or H_2O stress on LAI growth



Growth : Leaf area index

SBVMAX = maximum leaf area expansion per unit of accumulated biomass ($cm^2.g^{-1}$)

Maximal LAI growth rate = (DLTAMS + DLTAREMOBIL) x SBVMAX

Effect of trophic stress on LAI growth

Determinate crops

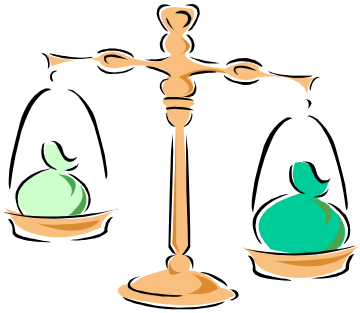
Trophic stress if radiations are limiting

Indeterminate crops

Sources :

Photosynthesis

Remobilizations



Sources/Sinks ratio

Sinks :

Fruits potential growth

Vegetative growth

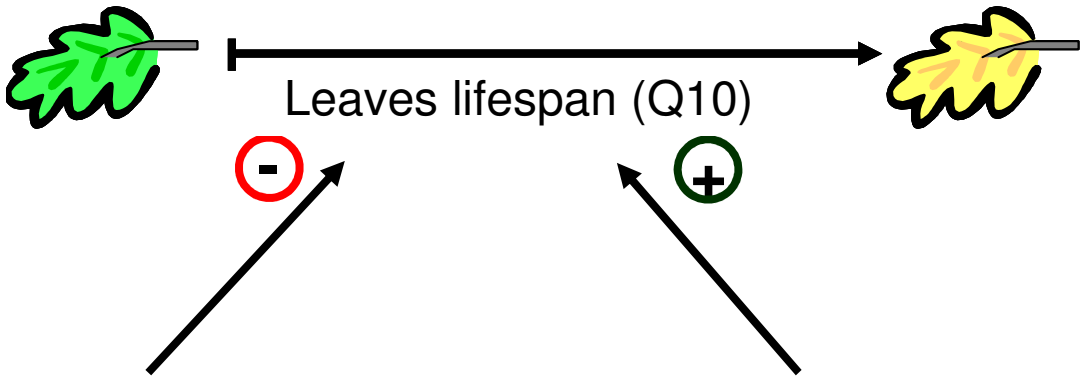
Growth : Leaf area index

Effect of senescence and stresses on LAI

Natural senescence

\+ Senescence is accelerated by stresses :
Water, N and frost

Senescence is delayed by over-fertilization



Stress N and/or H_2O

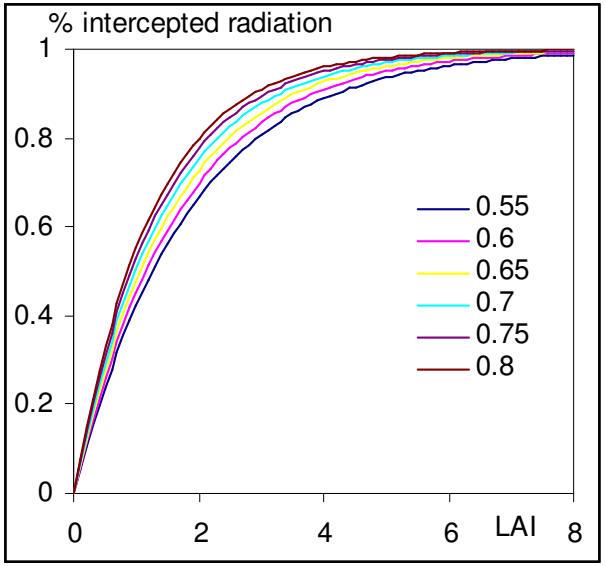
Over-fertilization N

Radiations interception

Growth : radiations interception

BEER law

intercepted radiation (%)



RADIATIVE TRANSFERS

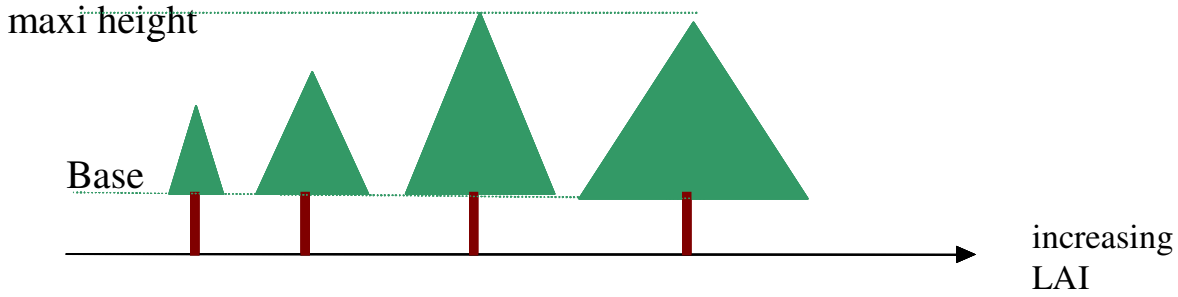
Choose a shape

leaf density profil types

Constant leaf density

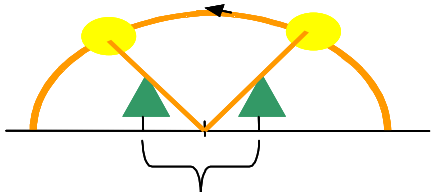
Width/thickness ratio constant until max height

Grow he shape

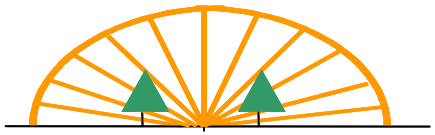


Daily radiation transfer

direct radiation



scattered radiation (47 directions)

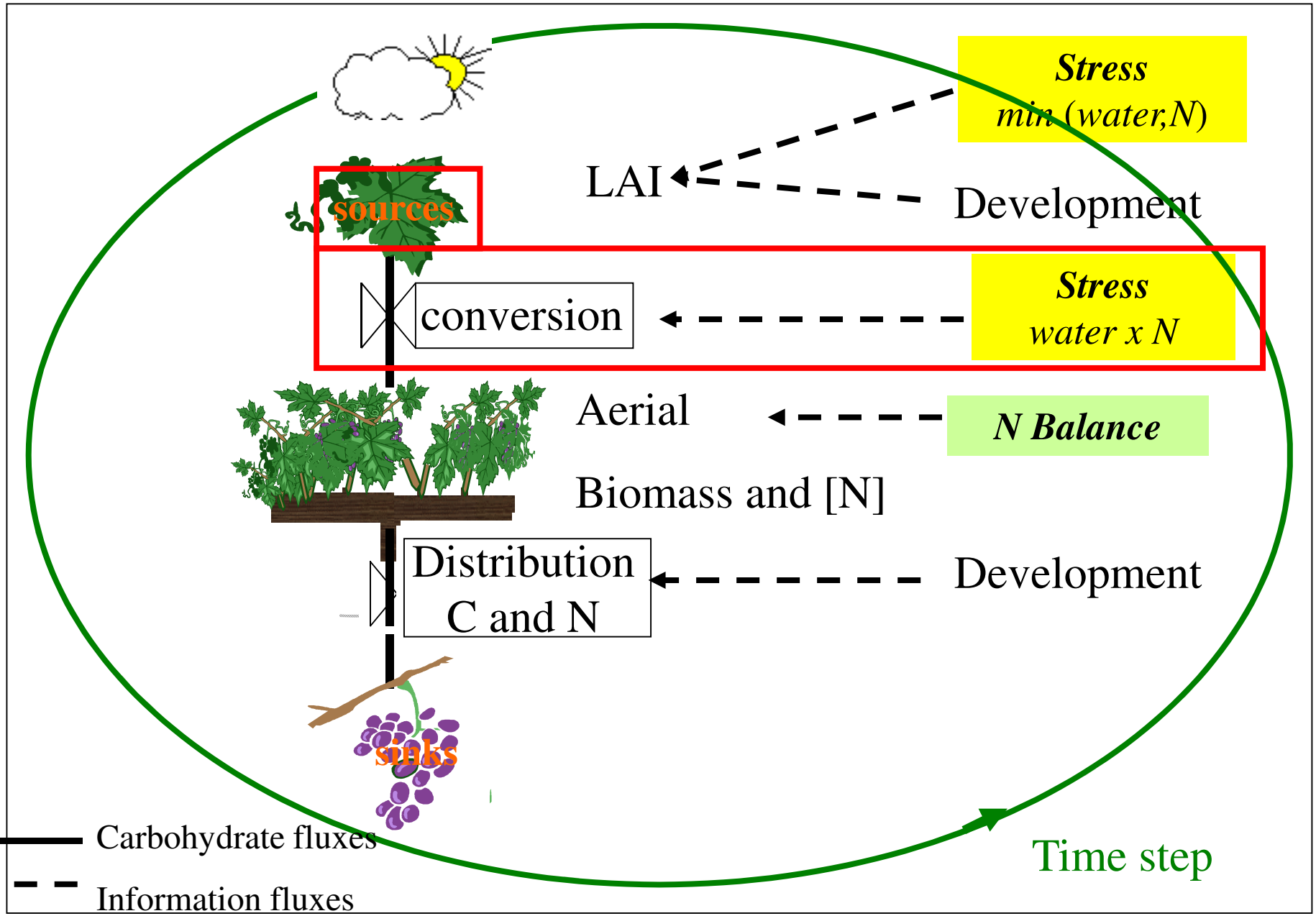


infinite crop

between-row sampling and mean

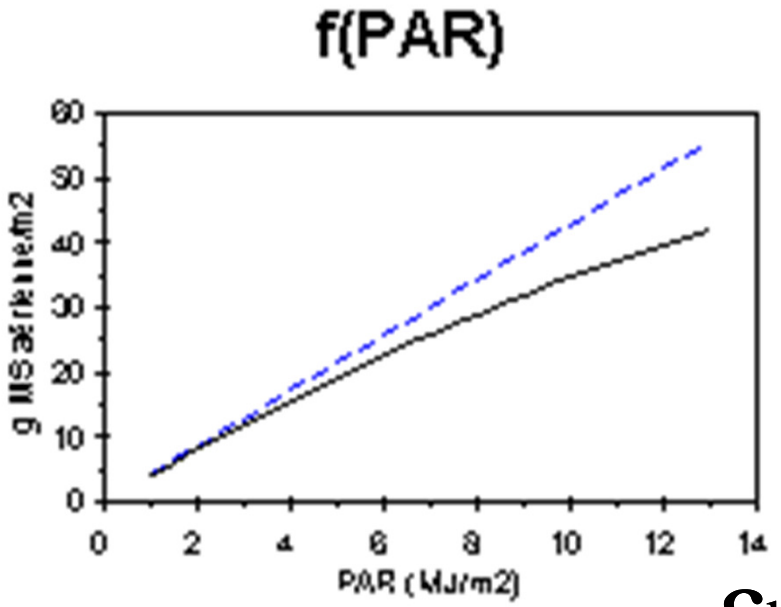
Biomass production

Growth : Biomass production



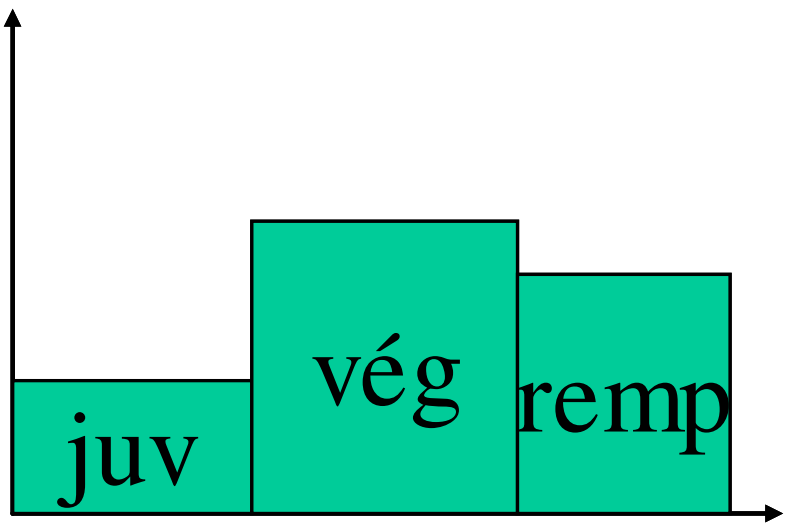
Growth : biomass production

Effect of radiations

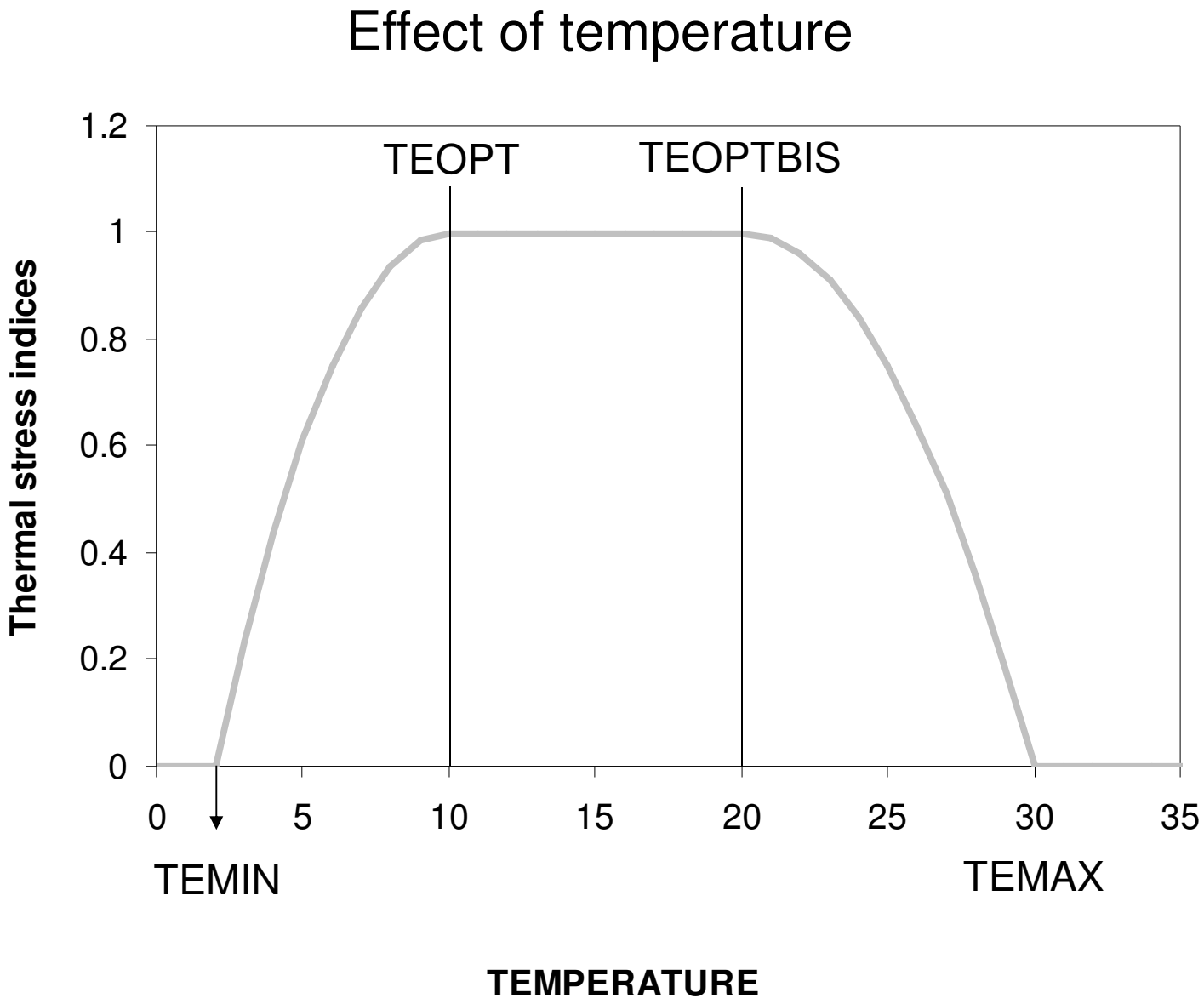


ε_{bmax}

Effect of development stage



Growth : biomass production



Growth : biomass production

Atmospheric CO_2 concentration affect

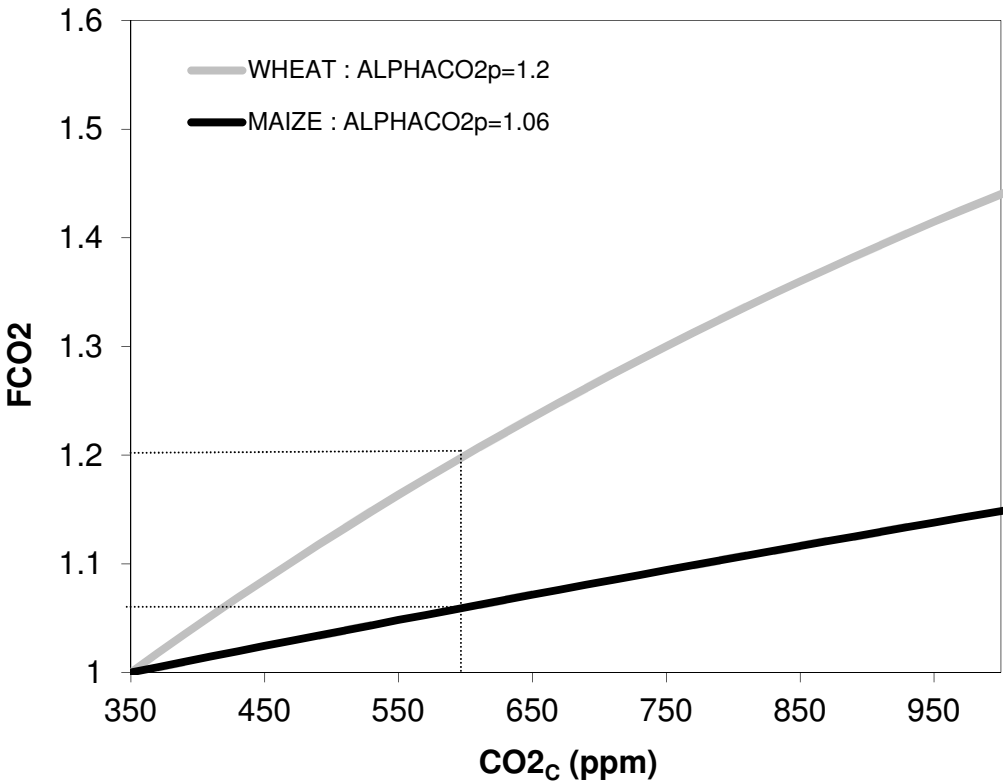
- Conversion efficiency
- Stomatal conductance

Conversion efficiency FCO2:

$$FCO2 = 2 - \exp\left[\log\left(2 - ALPHACO2_P\right) \cdot \frac{CO2_C - 350}{600 - 350}\right]$$

$ALPHACO2_P$
around 1.1 for C4 crops
around 1.2 for C3 crops

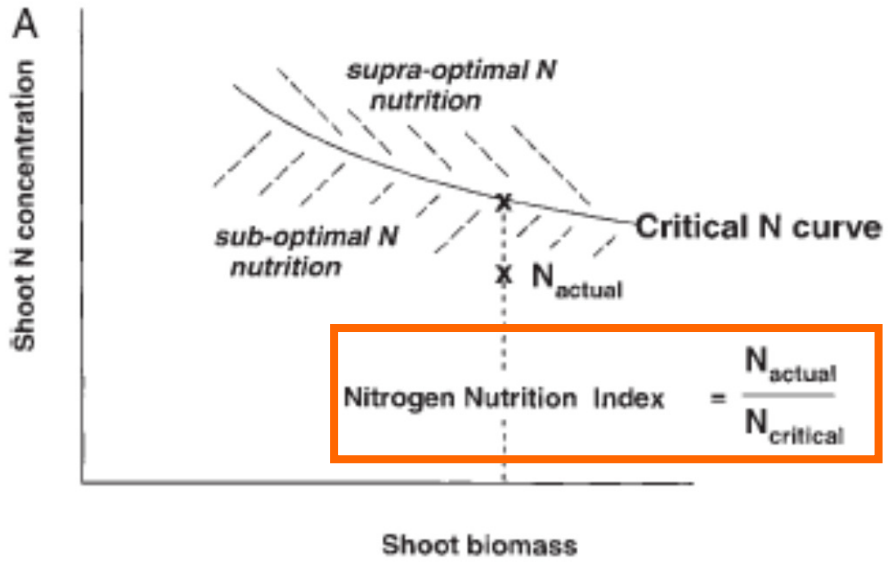
$CO2_C$
Atmospheric CO2
concentration (given daily)



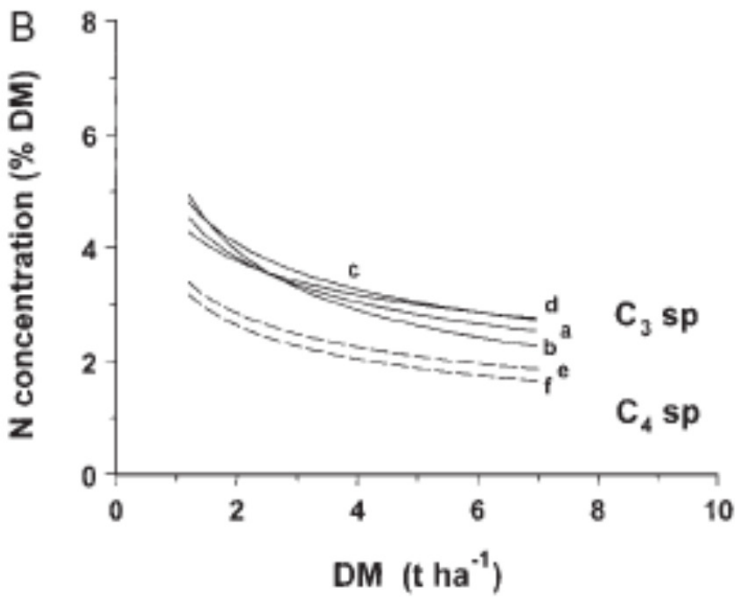
N stress

Growth : Crop and Nitrogen

Critical nitrogen dilution curve



- $[N]_{critique}$ = minimal N content to have maximal biomass production

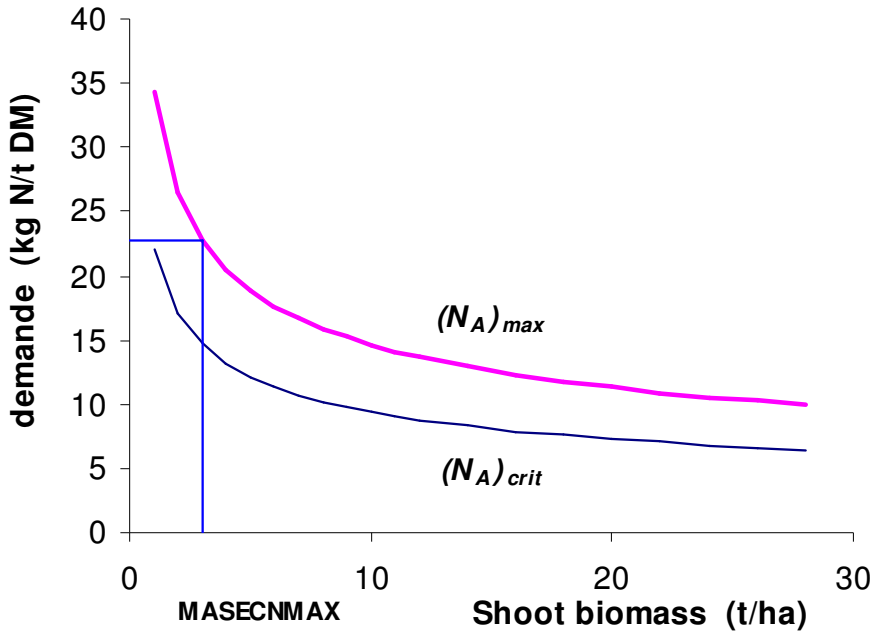


- C4 species have lower N need than C3 species

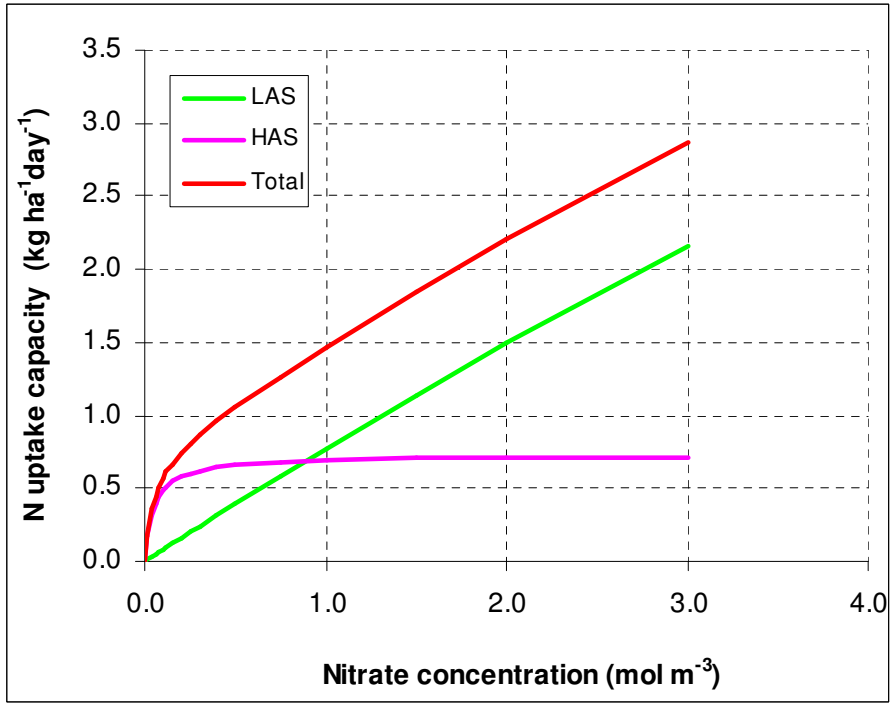
→ Due to a better N use efficiency

Growth : N uptake by the plant

Calculation of crop N demand

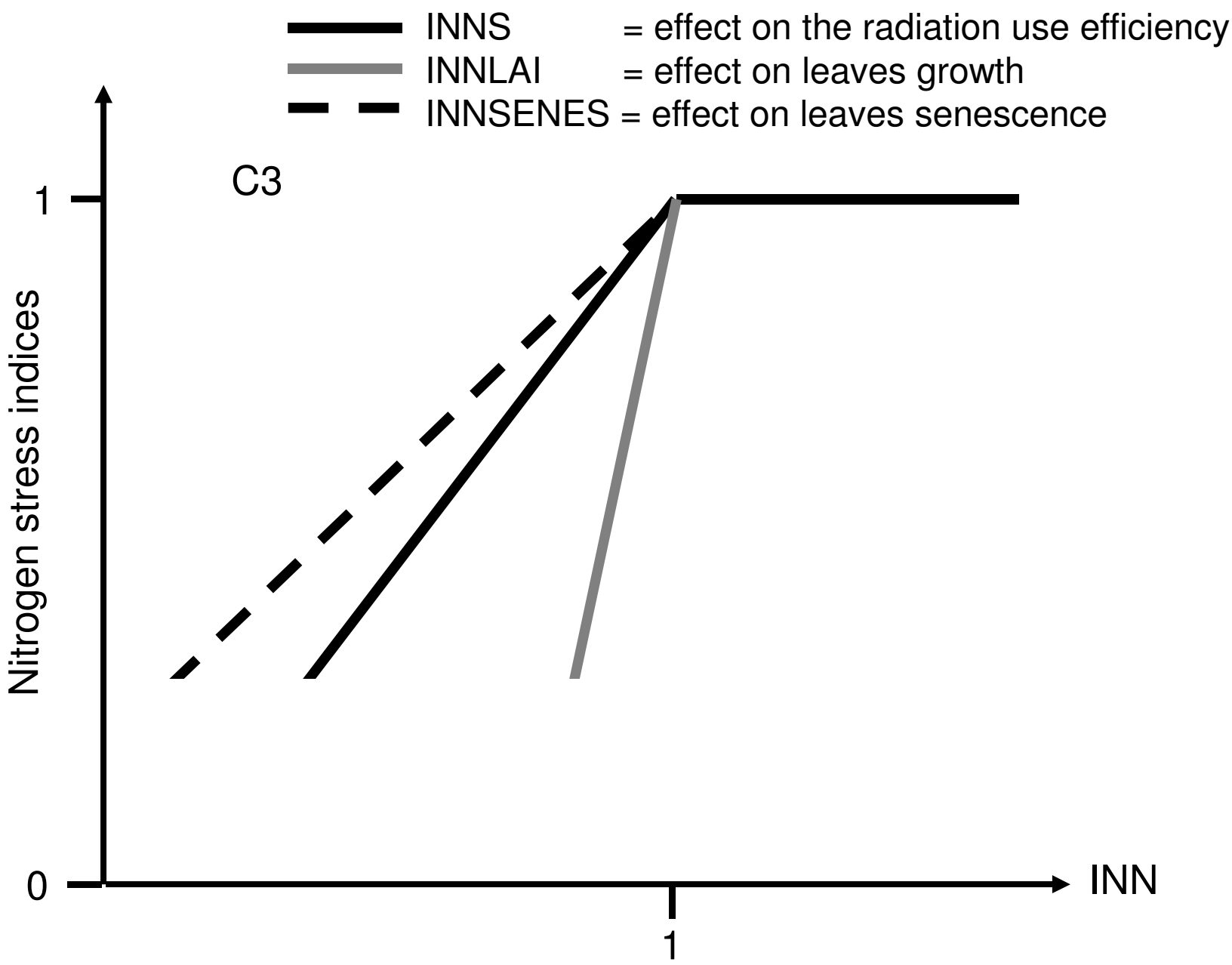


N uptake capacity

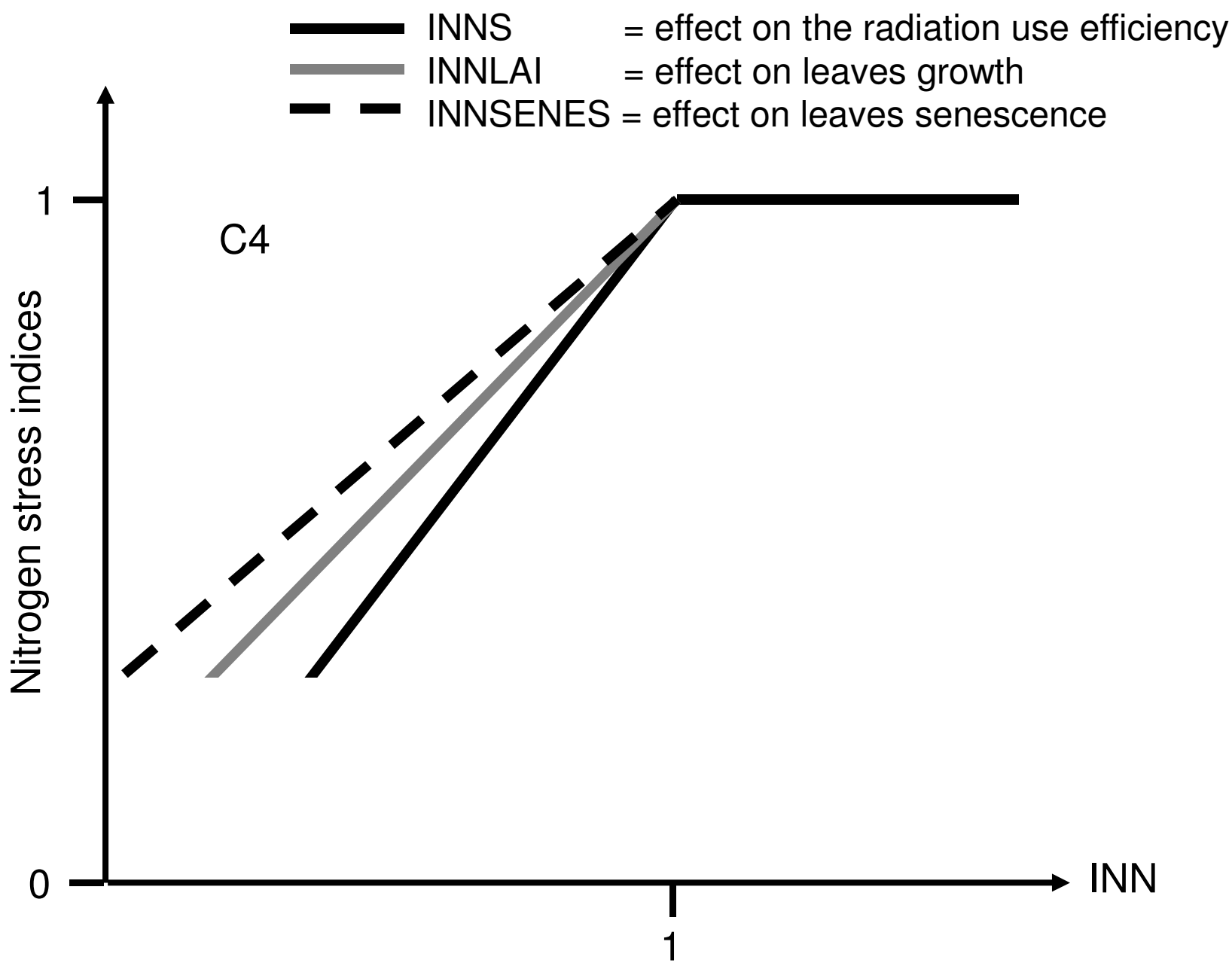


+ effect of roots density

Growth : nitrogen stress

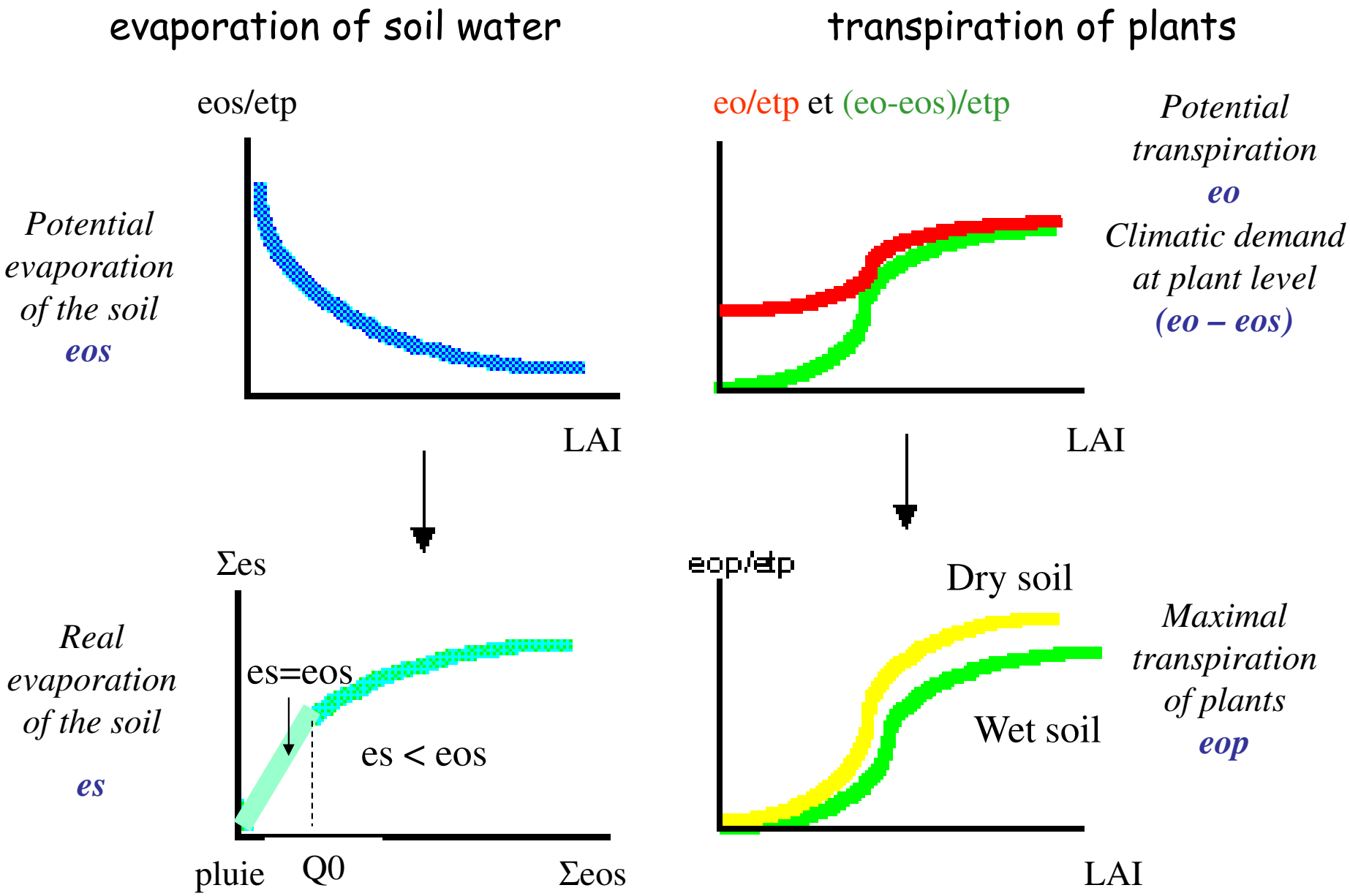


Growth : nitrogen stress



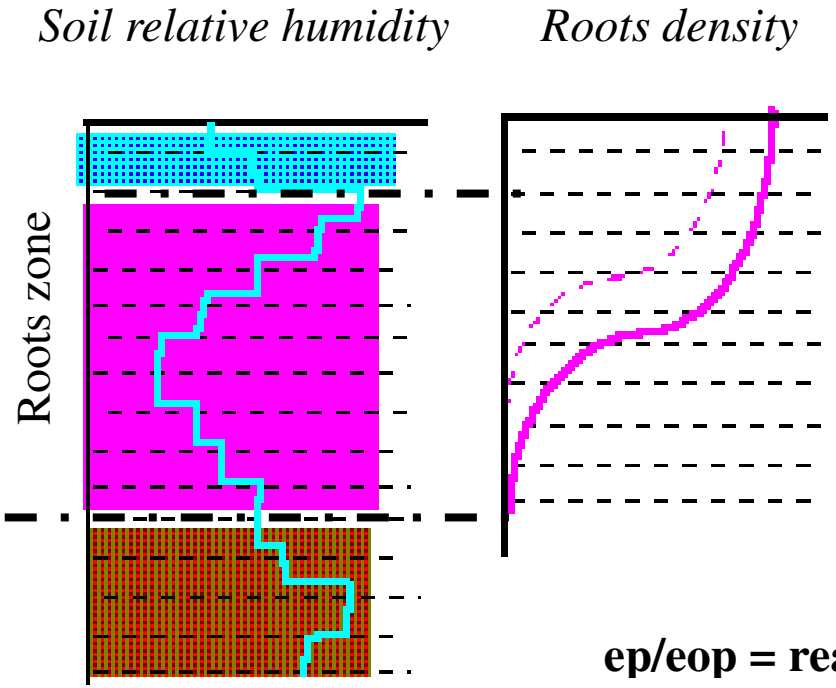
Water stress

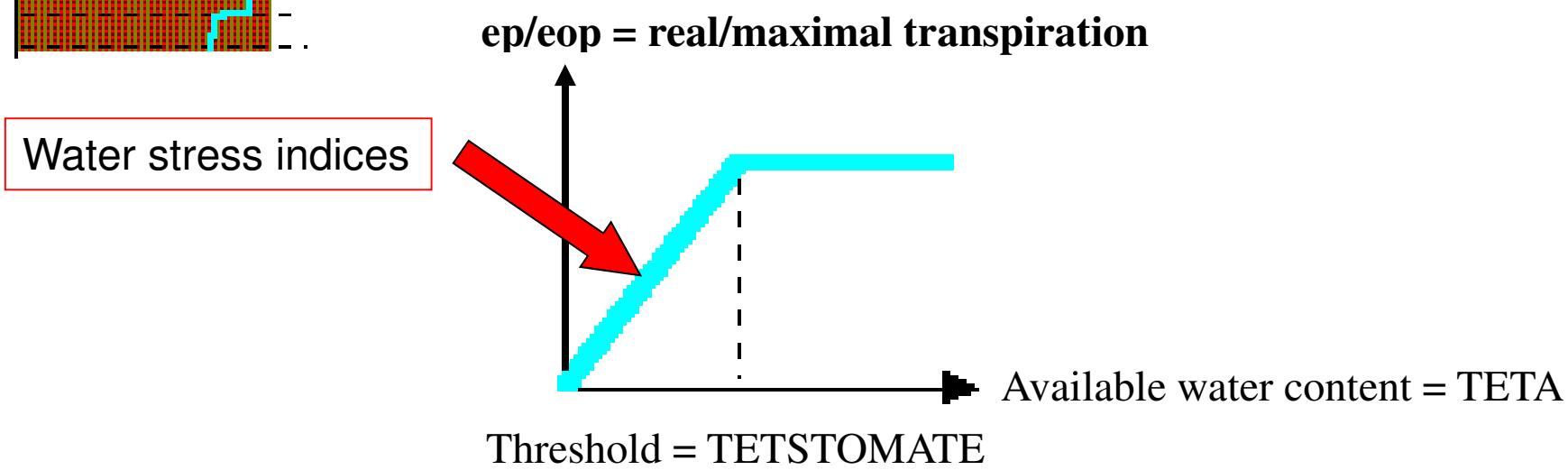
Growth : water stress



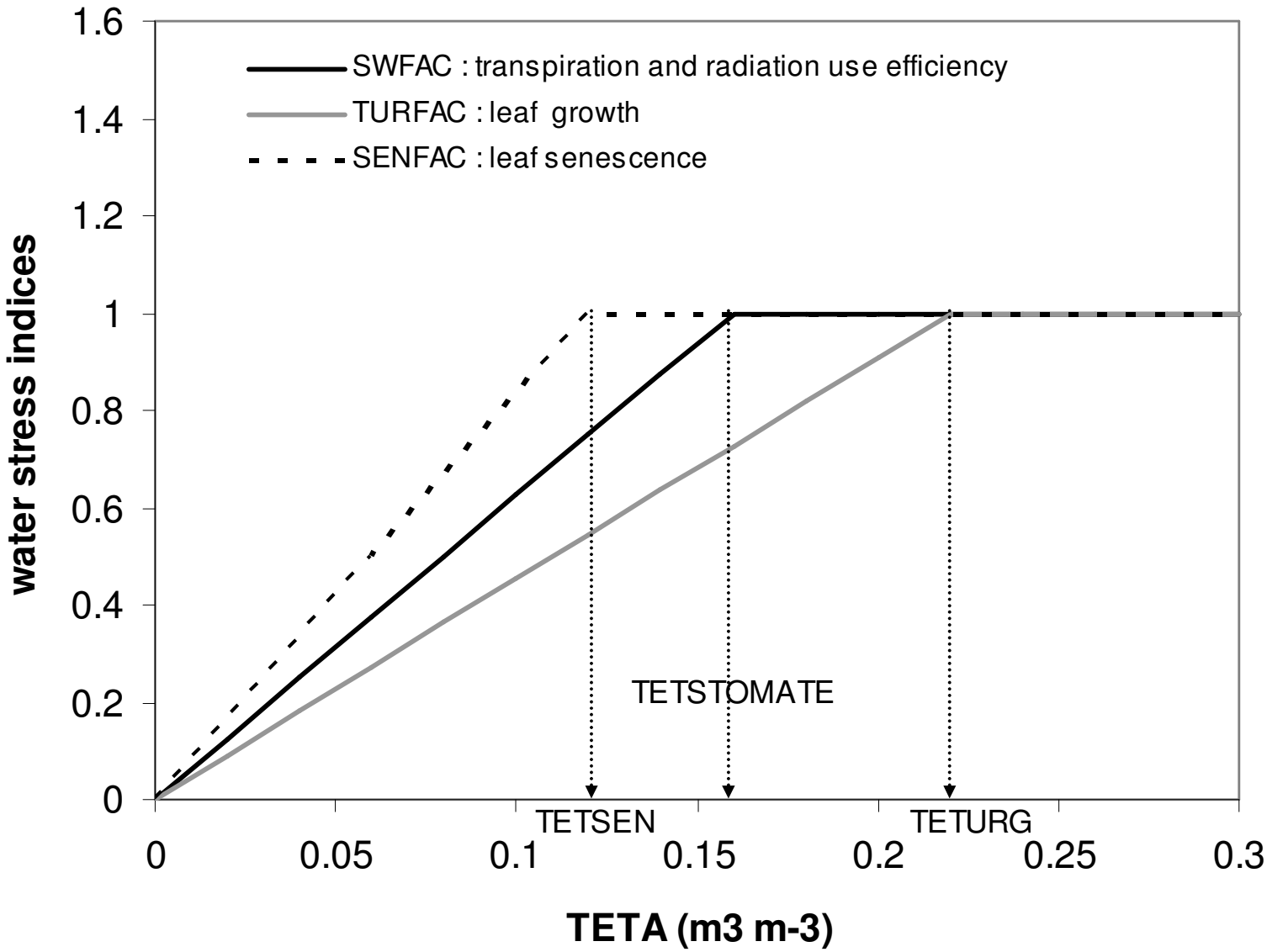
Growth : water stress

The real transpiration of the crop depends on 1) soil water content and 2) roots density



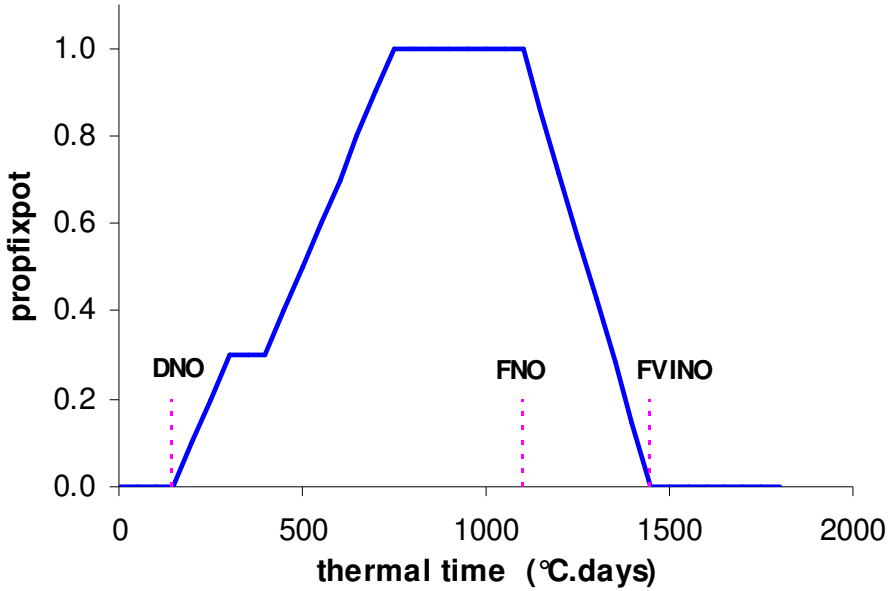


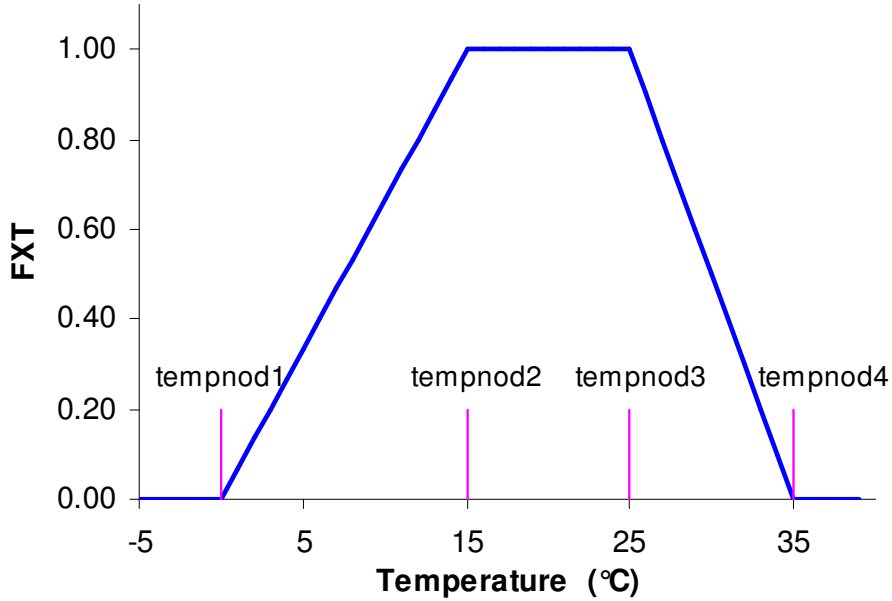
Growth : water stress

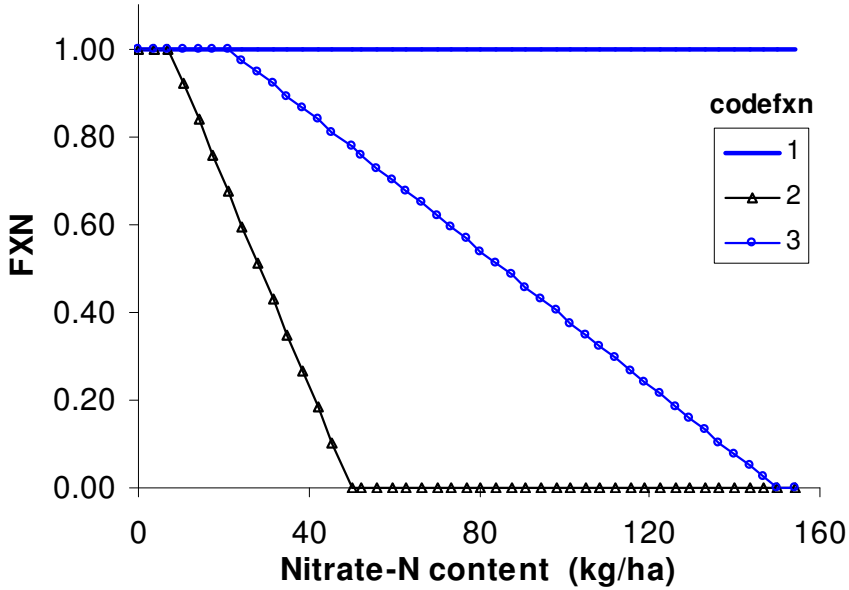


N fixation

Growth : the specific case of leguminous



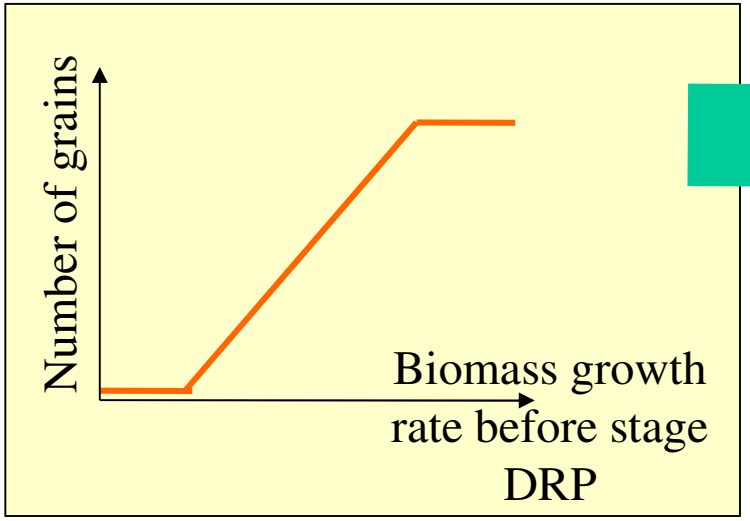


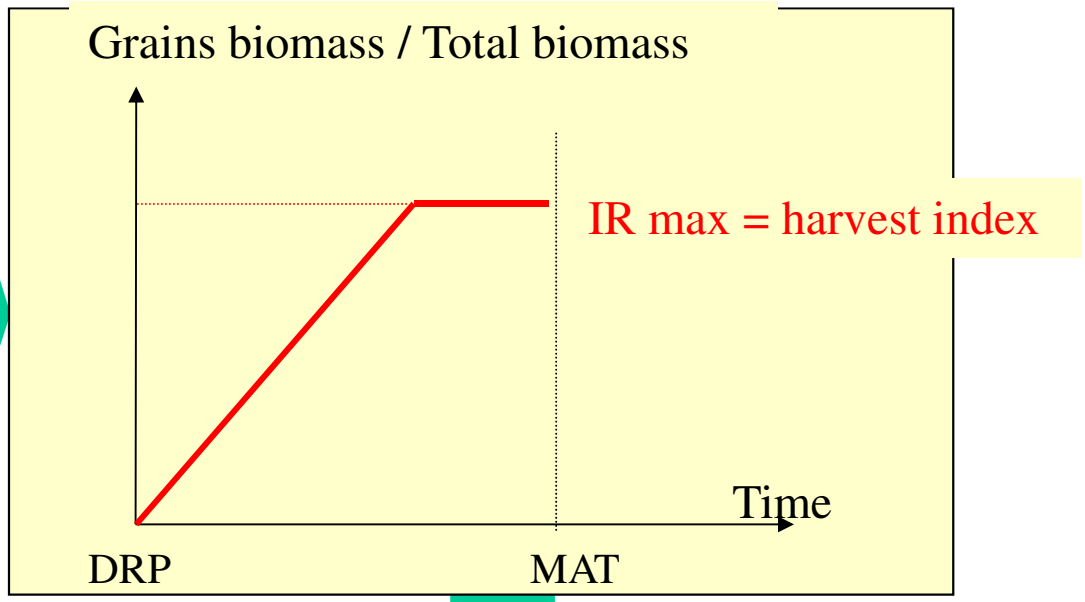


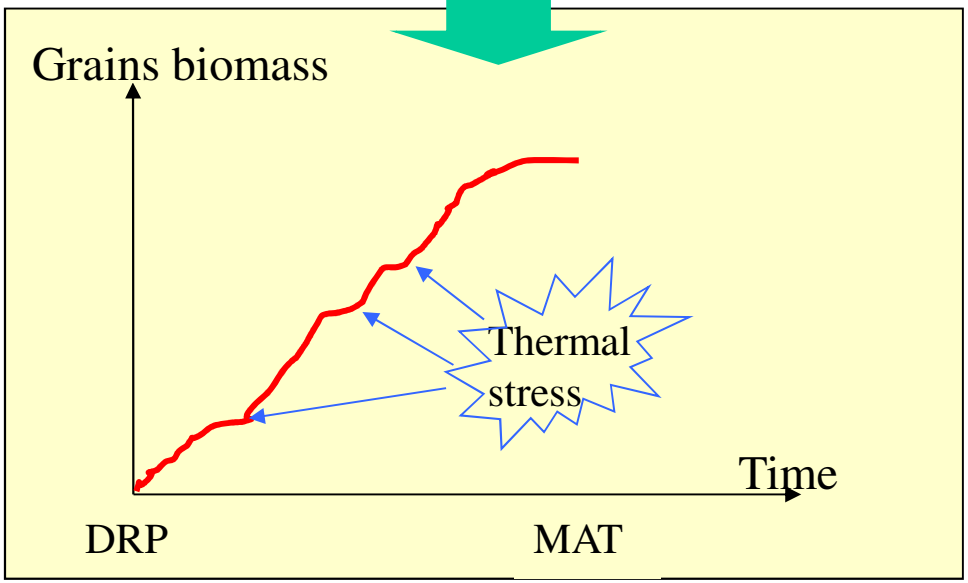
Yield

Growth : yield

For determinates growing plants



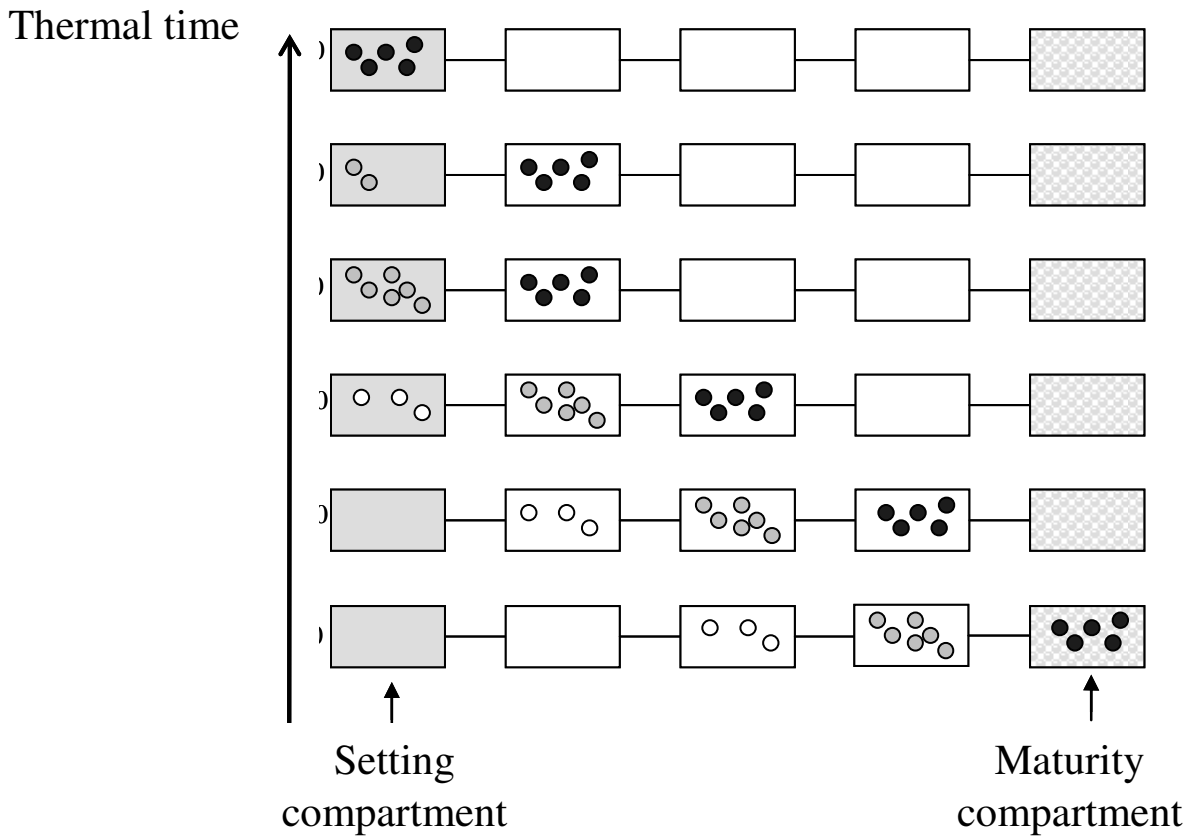




Growth : yield

<u>For indeterminate growing plants:</u>

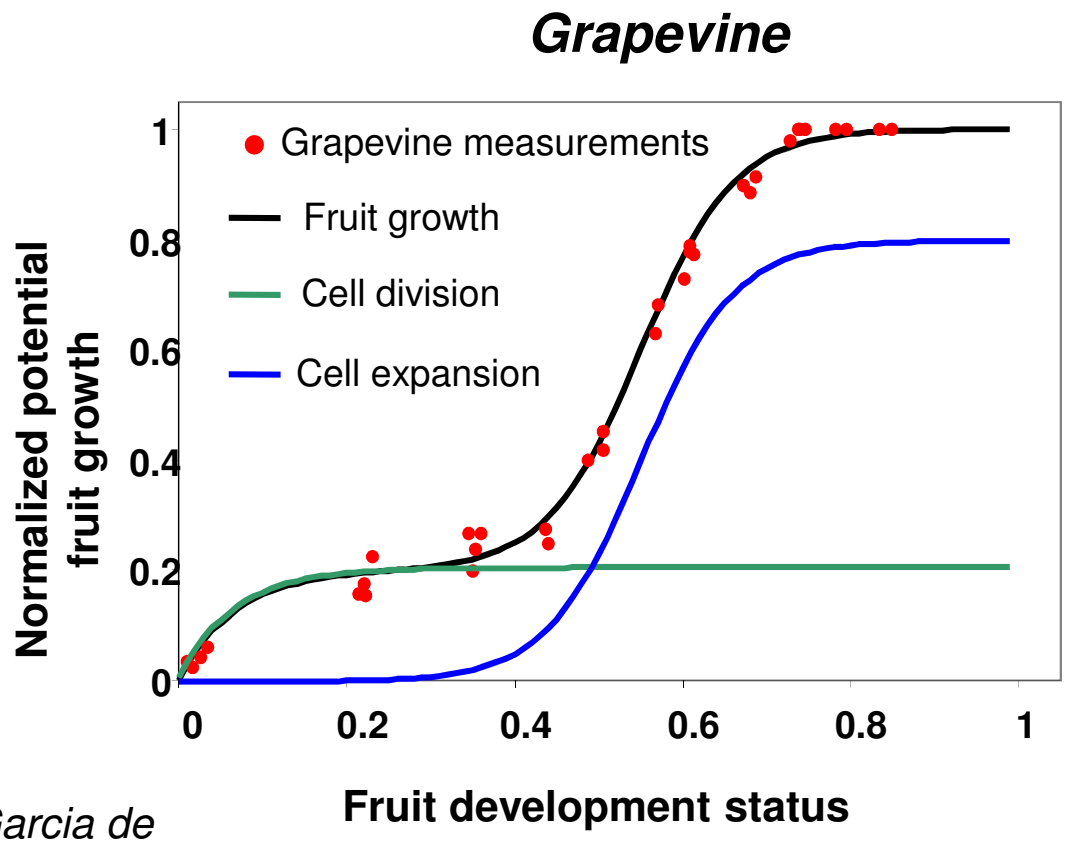
- trophic stress: sink/source approach
- dynamics of fruits cohorts: « boxcartrain » approach



Growth : yield

<u>For indeterminate growing plants:</u>

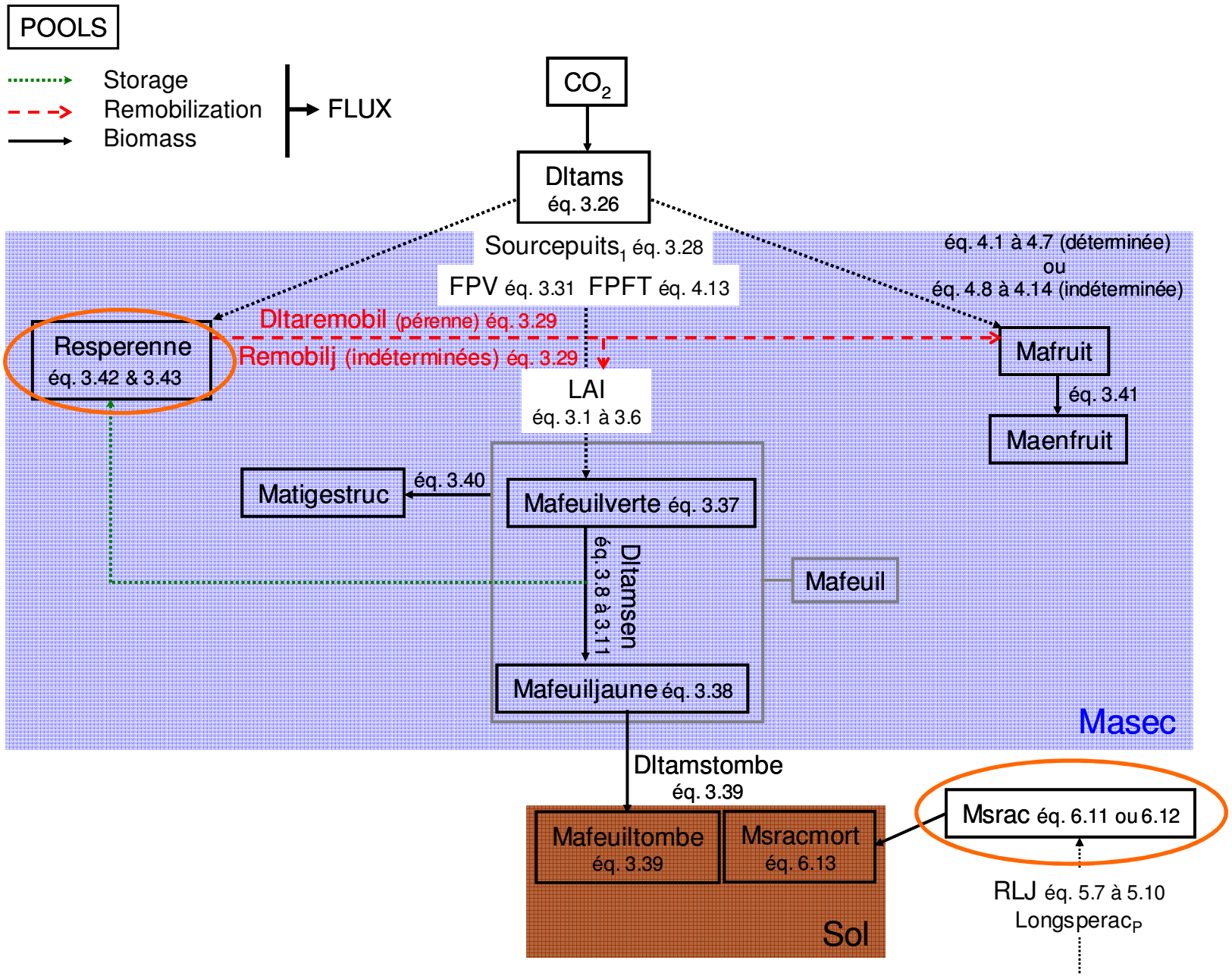
- trophic stress: sink/source approach
- dynamics of fruits cohorts: « boxcartrain » approach
- two successive phases in fruit growth: cell division, cell expansion



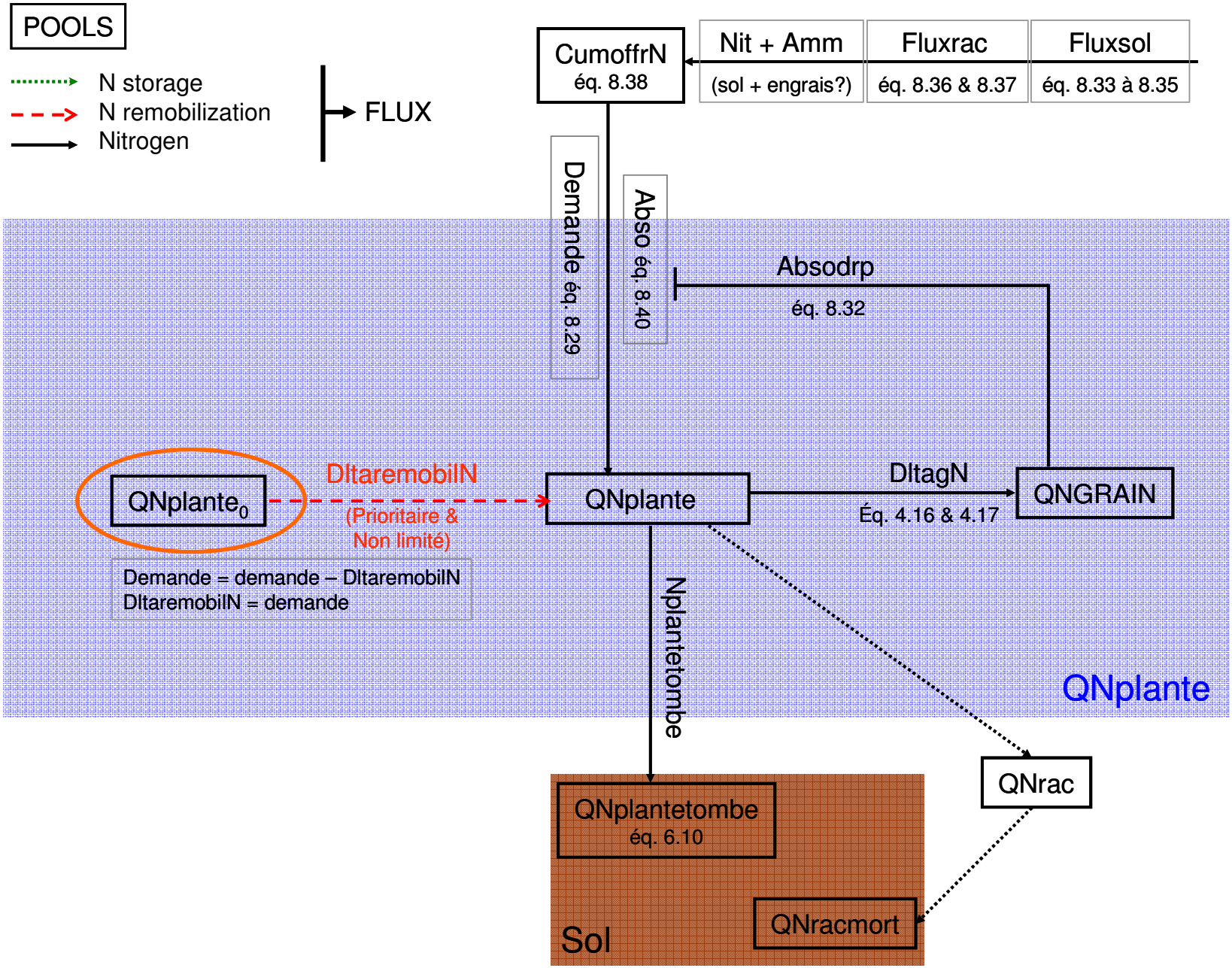
Garcia de Cortazar (2008)

Biomass and Nitrogen distribution

Growth : Biomass distribution



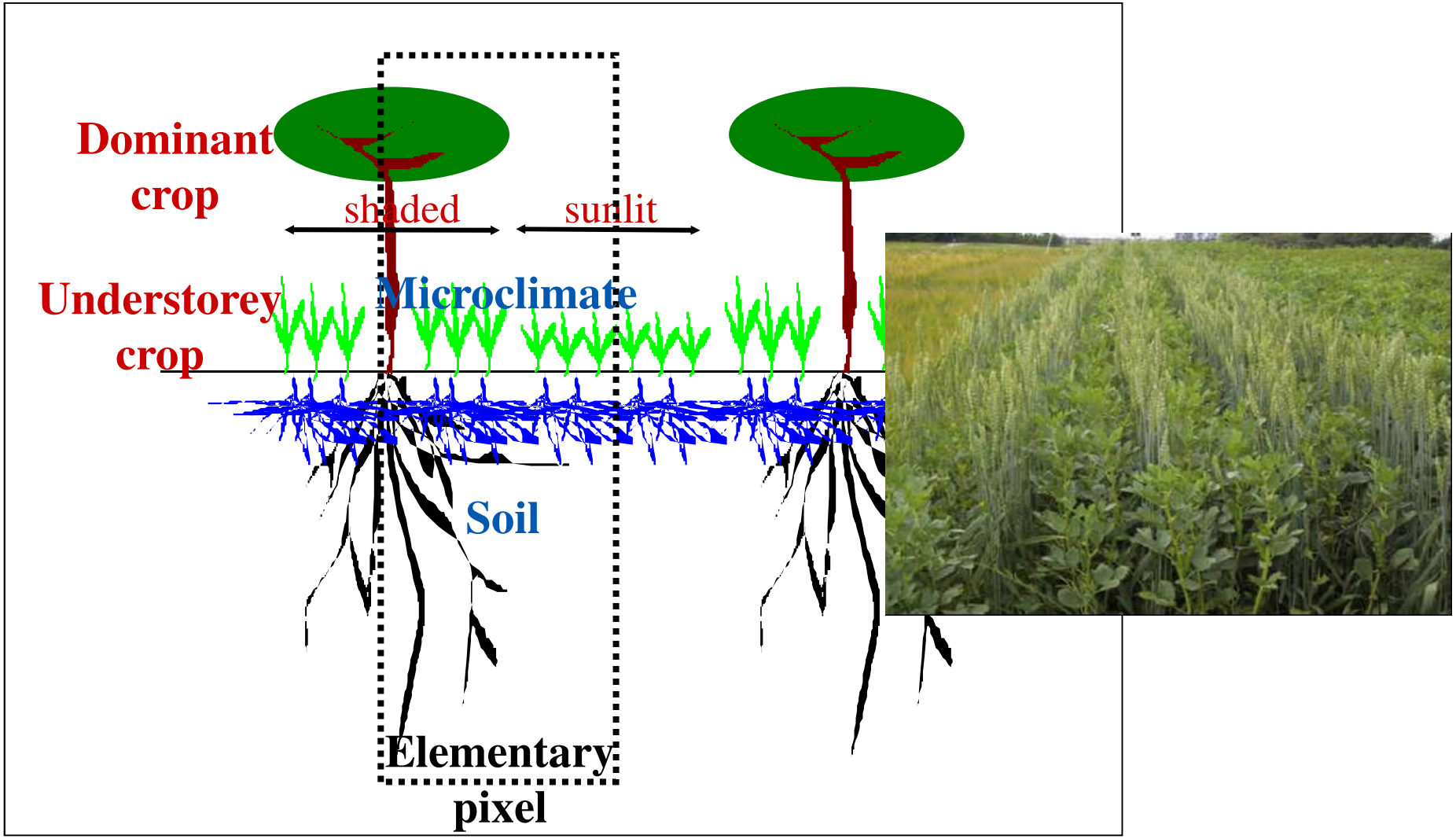
Growth : Nitrogen distribution



Intercropping systems

Intercropping systems

System

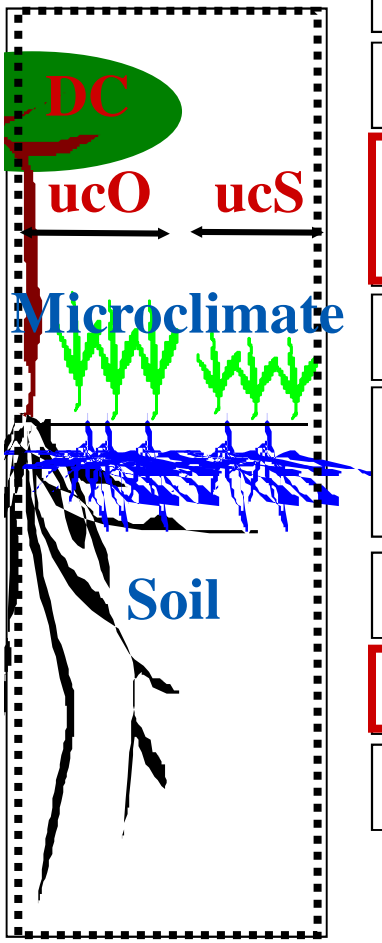


Intercropping systems

DC: Dominant crop

ucO: Shaded understorey crop

ucS: Sunlit understorey crop

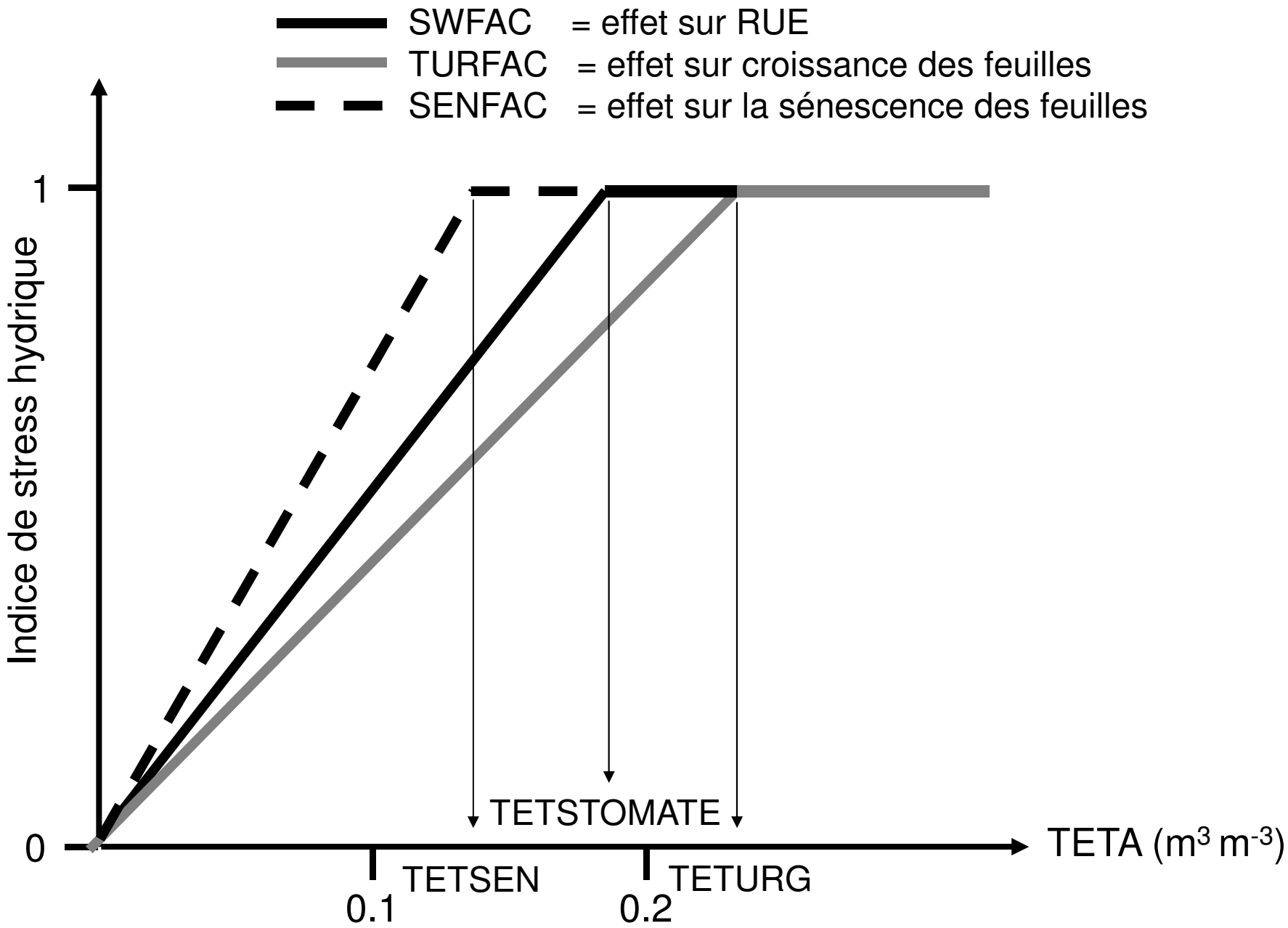


System	Process	Sub-system	
	Phenology	DC	uc
	Foliar surface growth	DC	ucO ucS
	Radiation interception	DC	ucO ucS
	Shoot growth	DC	ucO ucS
	Yield elaboration	DC	ucO ucS
	Root growth	DC	uc
	H_2O / N transfers	DC	ucO ucS
	Crop management	DC	uc

Modulo-Stics

Thank you for your attention

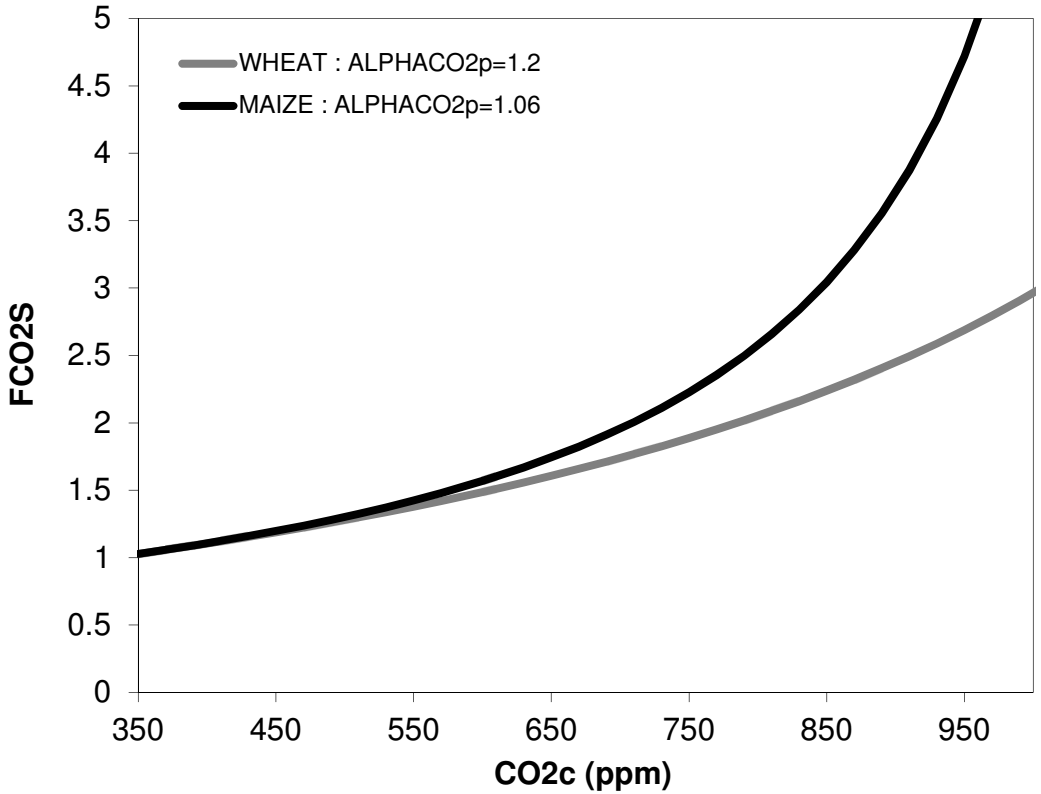
Module Croissance : Prise en compte du stress hydrique



Growth : biomass production

Stomatal conductance FCO2S:

$$FCO2S = \frac{1}{1 + 0.77\left(1 - \frac{FCO2}{2.5}\right)\left(1 - \frac{CO2c}{330}\right)}$$

FCO2
CO2 effect on conversion efficiency

$CO2_c$
Atmospheric CO2 concentration (given daily)